

Spatial mismatch and employment in Nigerian cities

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Abstract

This paper examines whether poor transportation infrastructure, proxied by city shape, impedes access to labor markets in large Nigerian cities. Combining novel primary data with Living Standards Measurement Surveys, I estimate the effect of city shape on labor force participation, employment type, and hours worked, constructing a potential developable footprint derived from each city's 1975 boundaries, historical population growth trend and natural geographic constraints to instrument for city shape. I find that more distorted city shapes are associated with higher commuting costs and greater public transit use, consistent with worse shapes imposing real transportation burdens on workers. IV estimates on labor market outcomes are mostly null but offer suggestive evidence in the LSMS sample that workers exit self-employment when shapes worsen, while city-level IV results indicate a hollowing-out of the firm size distribution in more distorted cities, with simultaneous increases in the share of one-person firms and large firms. Heterogeneity analysis reveals important nuances: worse shapes disproportionately reduce women's formal employment, with more pronounced patterns in smaller cities. LGA natives also respond differently from migrants, with worse shapes pushing them toward higher rates of formal employment. However, within-city variation in centrality and neighborhood quality explains little of the observed patterns.

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1 Introduction

African cities are growing at an unprecedented pace. Nigeria is the continent’s most populous country and a lot of the trends it experiences apply across the continent. Along Egypt, Ethiopia and Democratic Republic of Congo in Africa, Nigeria is one of seven countries that will contribute to half of the growth in the world’s city population over the next two decades (United Nations, Population Division 2025). Nigeria was only 20% urbanized in 1970 but is 63% urbanized in 2024 (United Nations, Population Division 2019, 2025). This rapid urbanization, however, has not been accompanied by commensurate investments in transportation infrastructure to match population growth and transportation demand. Poor transportation infrastructure may limit access to markets and public services for many residents of developing cities, with potentially severe consequences for urban labor markets.

The consequences of inadequate transportation infrastructure may extend far beyond inconvenience. In Nigerian cities, leapfrog development, where urban expansion occurs in disconnected patches rather than contiguously to existing development, is prevalent and is thought to increase infrastructure costs, limiting cities’ ability to build out infrastructure for utilities, transportation, and other essential services (Lall, Henderson, and Venables 2017) and preventing cities from benefiting from agglomeration (Glaeser 2008; Bryan, Glaeser, and Tsivanidis 2020). High transportation and commuting costs might make within-city moves either difficult or expensive, especially for low-income residents who lack access to private vehicles. In such a context, commuting to better jobs far from home can be particularly onerous, potentially trapping workers in low-productivity employment close to their residences or pushing them out of the labor force entirely.

Although transportation challenges facing Nigerian cities are stark, there is a paucity of data that can help document the issue. For example, data collected in the context of the UN’s 2030 Sustainable Development Goal¹ records that only 5.9% of the urban population in Nigeria had convenient access to public urban transportation in 2020, where convenient access is defined as living within five hundred meters of a low-capacity public transport system (bus/tram) or within a thousand meters of a high-capacity public transport system (Center For International Earth Science Information Network 2023) but the dataset only includes 23 cities covering approximately 24 million people, i.e. less than a quarter of Nigeria’s urban population, and only a minority of Nigeria’s major cities (Kano and Ibadan are for example absent from the list of cities). As of 2020, the motorization rate stood at just 56 registered motorized 4-wheel vehicles per 1,000 people in Nigeria, according to the International Organization of Motor Vehicle Manufacturers, far below the worldwide average of 209 vehicles. Across Sub-Saharan Africa more broadly, evidence suggests that the main and modal trans-

1. SDG 11 aims at “making cities and human settlements inclusive, safe, resilient and sustainable” (SDG Indicator 11.2.1).

portation method is walking (Kumar and Barrett 2008), and that commuting costs may be too high especially relative to low incomes (Kerr 2017; Kumar and Barrett 2008).

These limited transportation options may have important implications for labor markets. When commuting is onerous, workers might prefer working close to home, even for lower wages or less desirable employment or in low-productive sectors. This spatial dimension of job acceptance behavior may be particularly important in rapidly urbanizing African cities, where employment centers may be far from affordable housing and where public transportation systems are underdeveloped. In other low-and middle-income countries, especially in South Asia, transportation constraints have also been shown to reduce women’s labor force participation due to safety concerns, social norms restricting women’s mobility, and opportunity costs of time linked to domestic and care work responsibilities (Jayachandran 2021; Field and Vyborny 2022).

This paper addresses two main research questions. First, does lack of transportation infrastructure impede access to labor markets in Nigerian cities? Second, is there a difference in how different groups of workers (men versus women, local versus migrant) are affected by lack of infrastructure?

The empirical challenge in answering these questions is that transportation infrastructure is endogenous to economic outcomes. Cities with better labor market opportunities may attract more investment in infrastructure; unobserved factors such as governance quality or geographic features (sea shore, mountain range) may jointly affect infrastructure provision and labor markets’ performance. An additional challenge is that transportation data at a fine scale is rarely available for African cities, limiting what can be observed about actual commuting patterns. To address these challenges, I use city shape, defined as the geometric shape that the footprint of a city takes in space, as a proxy for transportation outcomes, reflecting both the increased cost of transportation provision and actual accessibility of the existing infrastructure. In urban geography, better city shapes (more compact, closer to circular) are associated with higher amenities, including better transit accessibility and better transportation infrastructure (Angel, Parent, and Civco 2010; Harari 2020; Duque et al. 2022). Elongated or fragmented cities may have longer average trip distances and require more extensive infrastructure investment to achieve the same level of accessibility as a better shaped city.

Following Harari (2020), I construct an instrumental variable for city shape that leverages each city’s boundaries in 1975 and geographic constraints surrounding the city (water bodies, steep slopes, wetlands) combined with projected linear population growth to predict how cities would have evolved absent endogenous planning decisions. Because this projected population is based solely on pre-1975 trends, it should not be affected by subsequent economic developments, infrastructure investments, or policy changes that might be correlated with both city shape and labor market outcomes. Intuitively, cities that are surrounded by

mountains or water or wetlands, all geographic features that raise the cost of providing infrastructure, will have more distorted and less compact potential shapes (and lower normalized shape metrics) because developable land will be fragmented due to these geographic features, whereas cities without these natural constraints would have more circular potential shapes as they can expand in all directions. This variation borne from natural constraints provides the source of identifying variation in the instrument.

The analysis relies on two datasets to examine these questions: the Living Standards Measurement Study (LSMS) for Nigeria in 2010 and 2023 that contains 51 of the largest cities in Nigeria (7136 observations) and a primary dataset I collected in 2025 covering 2462 individuals in thirteen major Nigerian cities. LSMS surveys are useful for identifying broader patterns and exploiting temporal variation in city shape, whereas the primary dataset provides rich data on commuting patterns, labor market choices, safety concerns during travel, and residential history that is not commonly available in standard household surveys for African countries. I complement these data sources with satellite imagery from the Global Human Settlement Layer to identify the built-up area footprints of urban settlements in Nigeria between 1975 and 2025. I then compute four shape metrics following Harari (2020) and Angel, Parent, and Civco (2010). Adapting Akbar et al. (2023b), I query Google Maps API to obtain estimated driving distance and duration and find that cities with worse shapes have longer average distances and travel times to the city center, validating the assumption that city shape metrics capture relevant variation in travel patterns and accessibility. The GHSL’s satellite-derived footprints and Google Maps validation allow me to confidently construct early city shape (used for the instrument) for a timeframe that predates the period of rapid urbanization in Nigeria, especially in the absence of historical and current data collected at the city level.

The findings present a nuanced picture. OLS results using the primary data show that worse city shapes, i.e. more elongated, fragmented, or distorted urban shapes, are associated with small increases in wage employment and hours worked. The OLS results using the LSMS sample differ from the primary data sample: moving towards worse shapes is associated with a statistically significant decrease in hours worked by 17, which is the opposite sign of the results in the 13-city sample. The opposite signs in OLS specifications for the two different samples suggest that they might be subject to different types of selection. The patterns derived from the OLS specification are still informative to think about transportation outcomes: worse city shapes are also associated with higher reported commuting costs and greater use of public transportation, consistent with the interpretation that distorted urban forms may impose higher transportation burdens on workers.

Moving to the instrumental variables approach, the IV results in the primary data sample show marginally significant decreases in formal employment and likelihood of working from home but all others are insignificant, reflecting limited statistical power in the 13-city sample rather than necessarily a true absence of effects despite a strong first stage. Relying on the

larger LSMS sample yields the finding that worse shapes affect self-employment (at the 10% level) as well as offer suggestive evidence of more hours worked, suggesting that workers who stay in the labor market in worse-shaped cities might work more to compensate for higher transportation costs. Overall, the sign reversal between OLS and IV is informative. Some of the worse-shaped cities might be able to compensate for shape in other ways (larger population centers, more dynamic labor markets, etc.). By contrast, cities for which these worse shapes inform their current spatial pattern and for which the instrument provides identification might be so distorted that economic activity has already been reorganized to account for the effect of bad shape. This could explain why the net employment effects are small: if workers in poorly shaped cities sorted into jobs close to home or already relocated to be closer to employment centers, the observable impact on employment outcomes may be modest even if transportation costs are substantially higher in these cities. As a whole, the LSMS IV results confirm that the labor market responses are muted, suggesting that workers may instead adjust their location choices or job search strategies to avoid prohibitively long commutes. This is consistent with Dean, Kreindler, and Mbonu (2026), who document significant dislike in traveling to unknown locations for urban casual workers in Nairobi, forgoing 22% of the median daily wages to avoid working in a location never visited before.

Perhaps the most striking finding is the limited evidence of differential effects by gender. Across the labor market outcomes examined (labor force participation, self-employment, wage employment, hours worked, and formal employment), men and women seem to respond similarly to city shape on most margins, as most of the interaction terms across all specifications are statistically insignificant. The exception is formal employment: women are relatively less likely to hold formal jobs when shapes worsen (statistically significant at the 1% level). I also explore differences by city size, showing that the net effect of worse shapes on many of these outcomes is large in small cities versus close to zero in larger cities. Nonetheless, I fail to reject equality of the female-shape interaction in small versus large cities in the sample. This stands in sharp contrast to findings from other low- and middle-income regions, particularly South Asia or Latin America, where transportation constraints have been shown to affect women’s labor force participation. In Pakistan for instance, Field and Vyborny (2022) finds that providing women with vouchers for safe transportation substantially increases their labor force participation, suggesting that safety concerns are a key barrier. In urban Egypt, Christensen and Osman (2021) shows that safety concerns in public transportation depress women’s travel demand. A review by Jayachandran (2021) documents the role of social norms such as restrictions on women’s mobility and stigma around women traveling alone in constraining women’s employment across low- and middle-income countries.

The limited gender differences in the Nigerian context may have implications for both research and policy. It suggests that the mechanisms linking transportation to women’s employment (social norms restricting mobility, safety concerns during travel, or even opportunity costs of time) may not operate in the same way in West Africa as in other regions.

Gendered social norms do not focus on female seclusion or on mobility restrictions; in this setting, only 11% of female respondents declare not being able to decide alone where they can go, with substantial variation across cities. 85% of women in the northern, predominantly Muslim city of Kano report this while only 1% of female residents in Lagos report it. Moreover, historically higher rates of female labor force participation can help explain why transportation might not be as big of a barrier for women in Nigeria and Sub-Saharan Africa in general compared to South Asia.

Moreover, safety concerns during commuting do not seem to differ as substantially by gender in the Nigerian context as in some Asian cities. In the primary data sample, women and men report similar levels of safety during commutes (2.42 versus 2.40 on a 7-point scale where 1 is very safe), and similar rates of experiencing harassment or assault during travel. This contrasts with contexts where fear of harassment or assault is a primary reason for women to avoid commuting or to restrict their labor market participation (Kondylis et al. 2020). These findings highlight the fact that interventions designed to improve women’s labor market access in African cities may need to focus on dimensions other than transportation infrastructure alone, or that transportation interventions should be designed differently than in other regions. While subsidized transport, women-only transport services, or improved safety on public transit have proven effective in South Asia and in Latin America, such policies may be less impactful in West African contexts.

As African cities continue their rapid expansion with the urban population projected to more than double by 2050, understanding how urban spatial organization affects economic outcomes becomes increasingly urgent. Yet models and interventions developed in other regions may not translate directly to the African context. Different economic structures (higher informality, more self-employment), different gender norms and roles, and urban forms (more fragmented growth, less formal planning, urban planning inherited from colonial times) prevail. The heterogeneous findings by gender and city size suggest a greater role as well of city size, with urban spatial organization in smaller cities mattering for gender equality.

This paper contributes to several strands of literature. First, it adds to growing evidence linking transportation infrastructure and spatial features of cities to labor market outcomes. Recent work has shown that urban compactness and spatial characteristics can explain wages and rents (Harari 2020), or infrastructure provision and downstream health outcomes (Baruah, Henderson, and Peng 2021). Expansions in public transit in Latin America has also led to reallocation of workers toward more productive firms (Tsivanidis 2019; Zárate 2019). This paper extends this literature to West Africa and explores city shape as a proxy measure of transportation accessibility in a context with limited data on urban transportation patterns. Secondly, this paper is among a handful of studies examining spatial mismatch and transportation in Sub-Saharan African labor markets. Residential segregation has been shown to contribute to high unemployment rates for black workers in South Africa

(Banerjee et al. 2008), aligning with more recent findings from the United States (Miller 2023). Providing fare subsidies to unemployed youth helps increase job search intensity and the probability of finding formal employment in Ethiopia (Franklin 2018) and South Africa (Banerjee and Sequeira 2023). However, most existing evidence comes from other regions of Africa, with limited work on West African cities where urban structure and labor markets may differ (considerably). Third, this paper contributes novel data on commuting patterns in Nigeria and examines whether commuting serves as an additional barrier to women’s access to the labor market.

Previous research has identified several mechanisms through which transportation can differentially affect women. Social norms may restrict women’s mobility or impose stigma on working far from home (Jayachandran 2021; Field and Vyborny 2022), while safety concerns could raise the psychological and physical costs of travel (Kondylis et al. 2020) and severely impact travel demand (Christensen and Osman 2021). Further, differential opportunity costs of time due to household responsibilities or dynamics (Picarelli 2019) may make long commutes particularly costly for women; this phenomenon of women’s relative disutility for commute time has so far only been demonstrated in high-income countries (Kwon 2022; Le Barbanchon, Rathelot, and Roulet 2021; Farré, Jofre-Monseny, and Torrecillas 2023).

2 Conceptual framework

City shape can determine transportation infrastructure (Harari 2020), affecting both the ease and cost of commuting across a city. In African cities, where housing markets are characterized by limited supply of formal housing and high transaction costs (Parby et al. 2015), workers might not be able to fully adjust by moving their home locations closer to firms. Absent adequate transportation, spatial frictions prevent workers from reaching labor market centers, generating high transportation costs that may be monetary or non-monetary: they include the direct monetary cost of transit fares or fuel, the cost of violence in the public space as measured by Kondylis et al. (2020), and the time cost of commuting.

Both time, monetary and safety cost considerations may limit the number of available jobs that a worker can commute to within a reasonable time frame, potentially disproportionately affecting low-income workers and women, for whom transportation costs represent a larger share of their budgets. Kerr (2017) shows that in South Africa, where commuting costs are large, they effectively function as a tax on earnings. The magnitude of these effects also depends on travel speed and the spatial distribution of jobs and city shape may affect travel speed by determining the length and directness of within-city routes². Christensen and Osman (2021) shows via simulation that improvements in travel speed within the Egyptian

2. Akbar et al. (2023a) finds that mobility in large Indian cities is impeded primarily by uncongested travel speed, rather than by congestion. Harari (2020) shows that compactness improves mobility in Indian cities.

public transportation system would generate substantial welfare gains. As discussed above, these costs are compounded for women through safety concerns and gender norms. Data on commuting costs and safety perceptions from the primary survey allow a direct test of whether transportation costs constitute the main channel through which city shape affects employment outcomes.

2.1 Labor supply model with commuting costs

I present a stylized static model of labor supply with commuting costs, drawing on Wang, Gong, and Wang (2020) and White (1988). Residential location is taken as fixed; workers commute to wherever workplaces are located and can change their commuting distance by changing jobs rather than by changing residence. Longer commute times are compensated by higher wages in equilibrium.

General set-up

A representative agent maximizes utility U , choosing between a consumption good x and leisure l : $\max_{x,l} U(x, l)$, where utility is increasing and concave in both arguments. The agent receives an hourly wage $w(d)$, with w increasing in d , and non-labor income y , with no savings. The numeraire good x has price 1, and both the wage $w(d)$ and commuting costs are expressed as functions of distance d , which denotes the round-trip distance from the worker's home to her workplace. The parameter t denotes commuting time per unit of distance, and c is the cost of commuting per unit of distance, encompassing both the direct monetary cost (fares, fuel) and any non-pecuniary costs evaluated in monetary terms, including safety concerns and time lost to (unreliable) transit. The worker allocates total available hours H across leisure l , work, and commuting time td . The budget constraint is:

$$x + cd = y + w(d)(H - l - td)$$

The maximization problem then becomes:

$$\max_{x,l} U(y + w(d)(H - l - td) - cd, l)$$

yielding the first-order condition, which characterizes optimal leisure and consumption on the intensive margin:

$$U_x(\cdot)/U_l(\cdot) = 1/w(d)$$

Assuming utility is equated across agents at the same residential location, so that each worker achieves the reservation utility $\bar{U}$, totally differentiating the first-order condition with respect to distance d yields:

$$w(d) = (1/t) [w'(d)(H - l - td) - c] \quad (1)$$

which implies the following wage gradient:

$$w'(d) = [w(d)t + c]/(H - l - td) \equiv [w(d)t + c]/n \quad (2)$$

where $n \equiv H - l - td$ denotes hours worked. The wage is increasing in distance ($w'(d) > 0$), and both a higher monetary commuting cost c and a higher time cost t reduce the wage offered at any given distance³. Working more hours ($n \uparrow$) raises the compensating wage. The signs of dn/dd and dn/dt are ambiguous, depending on the relative magnitudes of income and substitution effects; the sign of dn/dt is negative as long as the income effect is smaller in magnitude than the substitution effect.

Incorporating city shape City shape affects commuting by changing the speed at which workers can cover a given distance, independently of the city's actual area or size. Cities with fragmented or elongated shapes require longer detours and sustain less dense transit networks; however, properly developed transportation infrastructure can partially offset these effects of bad shape. Following Harari (2020), I parameterize commuting time per unit of distance and commuting distance as functions of shape S , where larger values of S indicate worse (less compact) city shapes, and $\kappa_t, \kappa_d, \lambda_t, \lambda_d > 0$:

$$t = \kappa_t + \lambda_t S$$

$$d = \kappa_d + \lambda_d S$$

The comparative statics below focus on the first equation, namely the effect of shape on commuting time per unit of distance, because in this framework distance is also a determinant of the compensating wage. City shape, by affecting distances within a city, may also raise reservation wages (to compensate for higher commuting costs) and increase job search costs⁴.

Incorporating gender The model also includes two sources of gender differences in commuting. First, the (monetary and non-monetary) commuting cost c may be higher for women, with $c_W > c_M$, reflecting greater safety risks and potential social costs of traveling alone for women (e.g. stigma). Second, women face a lower effective time endowment owing to domestic and care responsibilities, captured by $H_W < H_M$. These two differences help generate predictions about wages and hours worked, derived in the comparative statics that follow.

3. Formally, $dw(d)/dc = -t < 0$ and $dw/dt = -w'(d)(H - l)/t^2 < 0$.

4. As in Picarelli (2019) where housing proximity in the South African context shapes employment outcomes or in Franklin (2018) where transportation subsidies improve job search among unemployed Ethiopian youth.

Comparative statics and predictions

Prediction 1: City shape and labor supply Using the expression for hours worked $n = H - l - td$, substituting $t = \kappa_t + \lambda_t S$ and differentiating with respect to shape (holding distance fixed):

$$dn/dS = -\lambda_t d < 0 \quad (3)$$

Labor supply is lower in cities with worse shapes. This prediction is consistent with the literature linking transportation access improvements to higher labor supply in both low- and high-income countries. Holding d fixed, the expression in (3) does not depend on any gender-specific parameter. λ_t and λ_d can also vary by gender, showcasing the relative importance of shape in determining time and distance during commutes for men and women separately; in which case, (3) will depend on gender. In the general case, however, the slope of labor supply with respect to shape is the same for men and women, implying no differential gender response on the intensive margin when distance is fixed⁵. However, since $H_W < H_M$, the level of labor supply is uniformly lower for women:

$$n_W = H_W - l - td < H_M - l - td = n_M$$

Prediction 2: City shape and the gender wage gap Using equation (1), the wage at distance d is:

$$w(d) = [w'(d)(H - l - td) - c]/t$$

The gender wage gap at a given distance is:

$$w_M(d) - w_W(d) = [w'(d)(H_M - H_W) - (c_M - c_W)]/(\kappa_t + \lambda_t S) \quad (4)$$

Differentiating with respect to shape:

$$d(w_M - w_W)/dS = -[w'(d)(H_M - H_W) - (c_M - c_W)]\lambda_t/(\kappa_t + \lambda_t S)^2 < 0 \quad (5)$$

As city shape worsens ($S \uparrow$), the gender wage gap narrows. This occurs because worse shapes compress wages at all distances through higher t , and this compression is proportionally larger for men, who start from higher wages and higher effective hours. The result does not imply that women are better off in worse-shaped cities in absolute terms; rather, high-wage men experience a larger proportional wage penalty as shape worsens, conditional on the level of $H_M - H_W$ and $c_W - c_M$ ⁶. Note that this prediction on gender wage gap is

5. If shape also affects distance, the expression becomes $dn/dS = -\lambda_t(d + S \cdot dd/dS)$, which remains negative. The second derivative $d^2n/d^2S = -\lambda_t \cdot dd/dS < 0$, so the relationship is concave in this case.

6. One exception here concerns choice between formal and informal sector (not discussed in this model) with average commuting time t and distance d being lower for informal work (in some cases, $d = 0$ for

conditional on selection into employment remaining constant: if worse city shapes disproportionately drive low-productivity women out of the labor market, the observed gender wage gap could widen even as the model predicts narrowing among workers.

2.2 Connection to the empirical strategy

The two predictions above motivate the empirical design described in section 4. Prediction 1 implies that labor market outcomes should be worsening in city shape distortion, conditional on city size, with the effect operating through higher commuting time and monetary costs in more distorted cities. The main empirical challenge in estimating this relationship is that city shape is endogenous: cities with greater labor market opportunities may attract more infrastructure investment and more compact development, which would bias a naive estimate of the shape effect toward zero or positive values. To address this, I follow Harari (2020) and instrument city shape using the potential developable footprint, constructed from each city’s 1975 boundaries interacted with geographic constraints (water bodies, steep terrain, wetlands) and pre-1975 population growth trends, as described in section 4. The instrument predicts actual city shape through the mechanism in the model above: cities surrounded by geographic barriers develop more fragmented shapes as they grow, raising both t and c through the channels discussed above. The exclusion restriction requires that the instrument affects current labor market outcomes only through its effect on actual city shape, which is plausible given that it is constructed entirely from geographic features and population trends that predate the rapid urbanization of the sample period.

Prediction 2 motivates the gender heterogeneity analysis in section 5.3.1. The model predicts that, conditional on participation, the effect of city shape on hours worked should be similar in magnitude across men and women, since $dn/dS = -\lambda_t d$ does not depend on any gender-specific parameter when distance is held fixed, whereas level differences in labor supply may persist due to the fact that $H_W < H_M$ and $c_W > c_M$. The model does not necessarily predict a stronger response to city shape for women than for men, even as it predicts lower average labor supply for women. Whether this holds in the data is an open empirical question that the heterogeneity analysis aims to answer.

3 Data

3.1 City shape construction

Following Angel, Parent, and Civco (2010), Harari (2020), and Duque et al. (2022), I characterize city shapes using metrics that capture the geometric shape of a city’s urban footprint.

individuals working from their home locations), letting λ_t and λ_d vary by gender will also yield gender-specific predictions about reallocation across formal and informal sectors when city shape worsens.

In urban geography, better city shapes (more compact, closer to circle) are associated with higher amenities, including better transit accessibility and better transportation infrastructure. For each city polygon, I compute four shape metrics that capture different aspects of compactness and distortion: 1) Cohesion: The average Euclidean distance between any two points within the shape (taken from 20,000 interior points); 2) Proximity: The average distance from all interior points in the shape to the centroid of the shape; 3) Dispersion: The average distance from the centroid of the shape to 500 equally distributed points on the perimeter of the shape, 4) Spin: The average squared distance from all interior points to the centroid of the shape. Table 1 presents the proximity and cohesion metrics in 1975 and 2025 for all the cities in the primary data sample. Absolute metrics increase between 1975 and 2025 as cities expand in space, leading to more distorted shapes overall but normalizing by area in columns 1 and 2, shows that accounting for area, some cities actually become more regular in terms of shape as they grow.

One way to consider these metrics is to think of them as providing estimates of travel costs for within-city moves in the absence of reliable data, specifically proximity and cohesion most closely relate to commute-like trips within the city shape. In addition to reporting the individual metrics, I construct a summary index of the shape metrics by standardizing the first component of a principal component analysis of the four metrics. This index provides a single summary measure of overall shape distortion and is useful when presenting results compactly (its distribution is shown by year in Figure 2 and overall in A4).

These metrics are constructed by first delineating urban footprints from satellite data and then computing shape metrics for each footprint. I use the Global Human Settlement Layer (GHSL) Degree of Urbanization dataset, which provides 1-kilometer resolution classification of urban areas in five-year increments starting in 1975, which is the earliest year I use for city shape construction. The GHSL classifies each 1-km grid cell into categories based on population density and built-up area. To identify city shapes in 1975, I keep all pixels classified as “Urban centre”, “Dense urban cluster”, “Semi-dense urban cluster” or “Suburban or peri-urban” and exclude all pixels classified as rural and/or low-density. The union of all urban pixels within each city is converted into a vector polygon. For each city, I keep only the largest contiguous polygon from the union (not accounting for cities naturally separated by water bodies). This approach avoids complications from small, disconnected pixels that are not part of the urban core or may be included due to measurement error. The resulting polygon represents the spatial footprint of each city’s urban area in 1975. Figure A3 in the appendix shows city shape in 1975 for all cities in the primary data sample, while Figure 1 gives some indication of the variation in city shape relative to the mean shape, using the cohesion (average distance between any two points within the shape), proximity (average distance between the center of the shape and all points in the shape) and the summary index, going from better to worse moving from left to right. The same procedure is applied to construct city shapes for

Table 1: Proximity and Cohesion Indices by city, 1975 and 2025

City	Year	Normalised		Absolute	
		Proximity	Cohesion	Proximity	Cohesion
Abuja	1975	0.707	0.707	2.588	3.514
	2025	0.808	0.793	11.254	15.562
Benin City	1975	0.933	0.929	3.802	5.189
	2025	0.910	0.892	5.986	8.291
Calabar	1975	0.947	0.927	2.413	3.348
	2025	0.947	0.951	2.509	3.391
Enugu	1975	0.844	0.839	4.370	5.976
	2025	0.906	0.904	6.192	8.424
Ibadan	1975	0.923	0.912	7.191	9.883
	2025	0.955	0.946	10.459	14.343
Ife	1975	0.774	0.749	3.198	4.489
	2025	0.917	0.898	4.384	6.078
Ilorin	1975	0.890	0.882	4.393	6.024
	2025	0.930	0.914	6.514	9.002
Jos	1975	0.556	0.570	7.152	9.478
	2025	0.829	0.822	8.719	11.949
Kano	1975	0.870	0.845	6.588	9.216
	2025	0.938	0.920	9.804	13.582
Lagos	1975	0.869	0.847	11.641	16.219
	2025	0.713	0.681	19.380	27.562
Lokoja	1975	0.885	0.871	1.936	2.670
	2025	0.805	0.798	4.892	6.700
Ondo	1975	0.735	0.718	2.874	3.998
	2025	0.871	0.873	4.158	5.636
Oyo	1975	0.768	0.753	3.535	4.897
	2025	0.931	0.934	5.098	6.904

Note: Normalised indices are normalized by the corresponding value for a circle of equal area, so that a value of 1 indicates a perfectly circular shape and lower values indicate greater distortion. Absolute proximity and cohesion are in kilometres. Proximity index represents the average distance for each city between the centroid of the city shape and 20,000 interior points for each shape. Cohesion represents the average for each city between any two of the 20,000 interior points for each shape. Proximity and cohesion are expressed in kilometers.

2010 and 2025 using the corresponding GHSL data for those years. This allows me to track how city shapes have evolved over time, though the analysis relying on the primary data sample uses the 2025 shapes as the endogenous variable of interest and the analysis relying on LSMS links 2025 shapes to 2023-2024 LSMS data (and 2010 shapes to 2010 LSMS data).

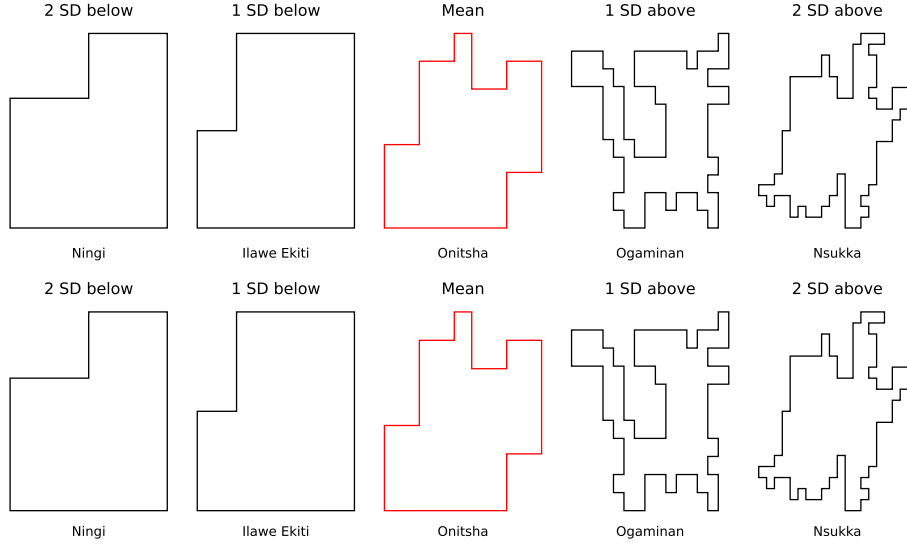


Figure 1: Cohesion (top) and proximity (bottom) indices in 1975: mean and standard deviation

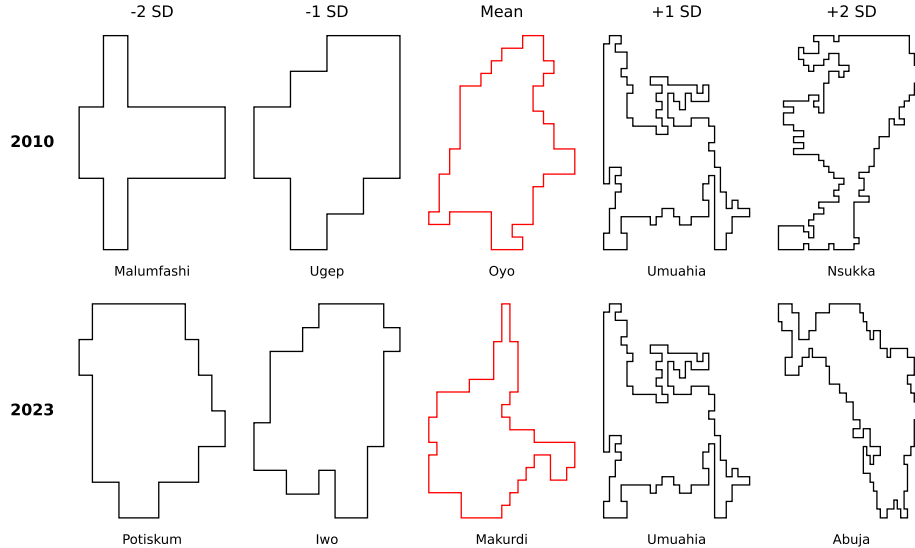


Figure 2: Summary index in 2010 (top) and 2023 (bottom): mean and standard deviation

To validate that city shape metrics actually capture relevant variation in travel patterns and accessibility, I adapt a procedure from Akbar et al. (2023b) using the Google Maps API.

This procedure can help provide external evidence that worse city shapes are associated with longer travel times and distances, supporting the interpretation of shape metrics as proxies for transportation accessibility. For each city in the larger sample, I randomly sample between 30 and 300 origin points (scaling down from $0.25\sqrt{\text{population}}$) within each city polygon, with sampling probability weighted by population (using gridded population data at the 500 meter grid cell). For all cities, I use the Google Places API to define the city center, that Google usually determines based on either the geometric center or a major civic landmark. For each origin-destination pair, I query the Google Maps API to obtain the estimated travel distance and travel duration by driving. I then compute the average distance and duration for trips for each city, and I relate these averages to the city shape metrics. Figure A1 confirms the expected relationship: cities with worse shapes (higher cohesion, proximity, dispersion, and spin) have longer average distances and travel times to the city center. In the figure, the cohesion index is normalized by the value of cohesion for a shape of same area, so lower values of the index indicate worse shapes. The relationship is negative and statistically significant for both distance and duration. This validates the hypothesis that the shape metrics capture real variation in accessibility and travel patterns, not just abstract geometric properties.

3.2 Dataset description

This paper relies on two complementary datasets: the Living Standards Measurement Study (LSMS) data and a primary data sample, both described in detail below.

The LSMS sample is composed of two rounds of LSMS data for Nigeria, conducted in 2010 and 2023. To identify relevant cities for the analysis, I identify all Nigerian cities with populations above 100,000 inhabitants in 2015, yielding a total of 51 cities⁷. The cities are indirectly identified in LSMS by matching them to the Local Government Area(s) they are composed of, as LSMS household survey data only identifies respondents down to this sub-administrative level. The cities are then matched to satellite imagery for city shape construction. The LSMS sample contains 7138 respondents across both rounds, aged 15 and above and residing in the 51 urban areas. The sample is approximately 51% female, and 63% of the respondents are employed in the past 7 days preceding the survey collection. The larger sample size, both in the number of cities captured and in the number of respondents, timeline of data available, provides substantial variation as well as allows comparison across time periods. However, for this study, as the LSMS was initially launched to examine rural livelihoods, it suffers from relatively low samples in urban areas (compared to the full sample

7. The initial cities were identified from the UN World Urbanization Prospects 2018 United Nations, Population Division (2019), which tracks all world cities with above 300,000 inhabitants in 2015. It contained as of 2019 50 cities for Nigeria. I lowered the threshold to 100,000 inhabitants to increase sample size relying on data from citypopulation.de and LGA population counts, reaching 120 cities in total. Out of those 120, only 105 are identified in satellite imagery and/or matched to pre-1975 historical population trends. 85 cities out of the 120 are found in the LSMS data in any year. Further, only cities that have at least five observations per year included in the analysis are included in the final analysis sample of 51 cities.

available) and some of the larger urban centers in Nigeria are not fully represented in it (only 44 cities out of 50 were initially matched in LSMS data). Moreover, geographic coordinates are jittered for confidentiality, limiting our ability to accurately assign households to specific neighborhoods within cities. Finally, the LSMS contains very limited data about commuting patterns. Respondents report whether they work but no information about commute mode, cost, or where they commute to is available. These limitations motivate the primary data collection detailed below.

The primary data comes from a survey I conducted during the spring of 2025 across thirteen major Nigerian cities: Abuja, Benin City, Calabar, Enugu, Ibadan, Ife, Ilorin, Jos, Kano, Lagos, Lokoja, Ondo, and Oyo (see Figure A2b). Together, these cities span Nigeria’s four major geographic regions of North, South, East, and West, including most of Nigeria’s major cities (Lagos, Kano, Ibadan, Abuja) as well as regional capital cities (Ondo, Oyo, Calabar, Enugu). According to the World Population Review, these cities represent a combined 40 million inhabitants, a combined 34% of the urban population of Nigeria. The sampling strategy I describe below was designed to survey multiple objectives, ensuring sufficient sample size within cities, geographic coverage at the country level, and cities across the 1975 city shape distribution, all within a limited budget.

Sampling of primary data sample I drew a random sample of twenty-four LGAs out of 109 LGAs, including all the major cities of more than 300,000 inhabitants in 2015 that comprised the initial sample. The sample was stratified to oversample LGAs from cities that were more distorted in 1975 based on the shape metrics described above. Since larger cities also have more LGAs, this strategy resulted in oversampling the largest cities.

Within each LGA, I sought to identify relevant neighborhoods for data collection, targeting low to middle income neighborhoods where a majority of residents would have employment and avoiding high to very-high income neighborhoods. Given the paucity of publicly available data at the neighborhood level, I developed an ad-hoc strategy using gridded satellite imagery and spatial data. More specifically, I computed grid averages for five variables (population density, population count, building count, building height and nighttime lights intensity), from publicly available satellite imagery, Gridded Population of the World, Google’s Open Buildings and VIIRS Nighttime respectively. I then used unsupervised machine learning approach (k-means clustering) to identify areas with high population density, low-to-medium nighttime lights and substantial building coverage but lower building heights. In addition to the clustering results I flagged all grid cells with population density above the 75th percentile, above median building counts, below average building height and night-lights⁸. I kept all grid cells that were selected with both of these selection mechanisms to randomly sample 8 and if there were not enough matching grids, I pooled remaining cells starting with the clustering algorithm than the statistical heuristic approach until I reached

8. I slightly relaxed these conditions to get to 7 cells by LGA.

8 grids (or “neighborhoods”) per LGA⁹.

Within each LGA, 100 respondents were interviewed, with no more than twenty respondents interviewed from the same neighborhood. This ensured geographic spread within LGAs and avoided having the sample dominated by a single area within each LGA. Respondents were sampled 60% of the time at their home, where only one respondent per household was interviewed, and the remaining 40% was sampled from transportation hubs¹⁰. Sampling at transportation was meant to help capture workers traveling to jobs away from homes and provide more variation in the information about commuting. The two types of sampling yielded similar respondents, with slightly lower incomes despite more hours worked, lower education and more out-of-state respondents (see Table A2) for respondents sampled in transit locations. To ensure gender balance, the enumerator alternated between men and women for each new interview. Crucially, the sample was restricted to respondents who were already active in the labor market, i.e. either had a job or were actively looking for one. This was a fail-safe to avoid capturing exclusively likely respondents staying home during workday hours, such as retirees or pensioners, students or long-term unemployed people, but precludes us from studying the extensive margin adjustments in labor market behavior.

The final primary data sample consists of 2462 respondents, with sample sizes ranging from 100 to 400 respondents per city, depending on the number of LGAs drawn within each city. The sample is 48% female and by construction 85% are active on the labor market at the time of the survey. The primary survey collected detailed information across multiple topics not typically available in standard household surveys (see detailed description in Appendix A.1). The labor market module was created to hew as closely as possible to the LSMS employment module. The main labor market variables are an indicator for being active on the labor market, employment status (employed, self-employed, unemployed), hours worked in the past week on the main job, (number of months worked in the past year), and an indicator for formal employment. The survey contains novel data on commuting in Nigerian cities. Respondents reported their exact home location with GPS coordinates and address and work location, transportation assets, usual commute time, cost, and primary mode of transportation to commute to work. Finally, the survey also contains a detailed residential history and location choice module, that includes questions on neighborhood’s access to public infrastructure facilities, and a module on feelings and perception of safety in their

9. The resulting list was manually reviewed at two points in the process to drop inhabitable pixels, industrial areas or other non-residential areas. Due to widespread safety concerns in Nigeria, enumerators were asked to individually review the list they received and flag any potential security issue in any of the eight areas for each LGA that they felt could pose a concern for their safety. After review by the research manager, areas that were then deemed too dangerous for an enumerator to survey alone were dropped. Data collection in Mushin LGA in Lagos was stopped after the enumerator was mugged, and an additional LGA was resampled from the remaining Lagos LGAs.

10. The transportation hubs were sampled from lists for each city/LGA created using the Google Maps Places API, with ex-post manual correction for duplicate listings or erroneous location information.

neighborhood and during commutes, as well as experience with crime and traffic accidents, and reasons for choosing their residential location. This rich data allow me to examine not just whether city shape affects employment outcomes, but also the mechanisms through which these effects might operate.

Beyond the labor market surveys and city shape data, the analysis draws on several additional sources. Historical and projected population for Nigerian cities from the UN World Urbanization Prospects 2018 and citypopulation.de, disaggregated at the LGA-level. These data are used to project city growth for the instrumental variable construction (described in the next section). Geographic constraints data is constructed from MODIS Raster Water Mask and the JRC Global Surface Water to identify pixels covered by any water body such as lakes, rivers, coastline or wetlands, as well as Shuttle Radar Topography Mission (SRTM) data to compute terrain slope and elevation. These data sources are used to identify developable land pixels for the instrumental variable, as described in the next section. City-level manufacturing shares are reconstructed from LSMS data. These data are used as controls in robustness checks to address concerns that city shape may be correlated with city-level manufacturing share.

3.3 Summary statistics

Table 2 presents summary statistics for the primary data sample. The average respondent is 35.7 years old and has completed 13.0 years of education. About 90% can read and write in English. Approximately 69% of respondents live in their state of birth, while 39% live in their LGA of birth, indicating substantial migration within Nigeria. About 48% of respondents self-report living in an informal settlement. Regarding labor market outcomes, 85% of respondents are active on the labor market, 6% are searching for a job, 43% are self-employed, and 25% are wage employed. Respondents work an average of 36 hours per week. About 20% work in formal jobs (defined as jobs with written contracts or having a health insurance scheme or a pension). Average weekly income is 24,585 Naira (approximately \$16 as of November 2025 exchange rates), which is about 52% above the Nigerian minimum wage of 70,000 Naira per month¹¹. About 29% of respondents work from the same location as their home, which is more common among the self-employed (35% versus 25% of the non self-employed).

Overall, the primary data sample is comparable to the LSMS sample shown in Table A1, which is on average a bit older, less educated and potentially poorer (lower reported monthly wage, lower hours worked and less likely to live in the largest cities of the sample). Comparing the population data in the two samples, the primary data sample contains only large cities that had at least 300,000 inhabitants by 2015, whereas the LSMS sample contains a more balanced distribution in city size, with 27% of the sample living

11. According to the National Minimum Wage (Amendment) Act 2024 signed in July 2024

in cities that had 300,000 inhabitants or less by 2015, 16% between 300,000 and half a million inhabitants, 20% in cities with half a million to a million inhabitants and the remainder (32%) living in cities with more than a million people. 13% of the sample comes from Lagos. The primary data sample had 45% of the sample coming from cities between 300,000 and 500,000 inhabitants, 12% from cities with less than a million people and the remainder (42%) coming from cities above a million people, including 17% from Lagos alone.

Table 2: primary data sample description

	Male	Female	All	<i>p-value</i>
Age	36.50	34.92	35.73	0.000***
Can read and write in English	0.89	0.91	0.90	0.189
Completed years of education	12.95	12.96	12.95	0.969
Lives in state of birth	0.69	0.68	0.69	0.369
Lives in LGA of birth	0.43	0.36	0.39	0.001***
Lives in an informal settlement	0.51	0.46	0.48	0.026**
Feels safe (1 to 7)	2.38	2.40	2.39	0.547
Active on labor market this week	0.83	0.87	0.85	0.004***
Searching for a job this week	0.06	0.05	0.06	0.378
Self-employed	0.38	0.48	0.43	0.000***
Wage employed	0.25	0.25	0.25	0.626
Number of hours worked in past week	35.25	36.90	36.04	0.172
Formal job	0.22	0.19	0.20	0.042**
Average weekly income	25834.97	23027.18	24585.45	0.135
Average weekly revenue (if self-employed)	33531.30	26558.19	29672.39	0.070*
Work is same location as home	0.27	0.32	0.29	0.010**

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column 1 and 2 report the sample means for male and female respondents respectively, column 3 reports the combined sample means and column 4 calculates the p-value for the t-test of the difference in means column 2 and column 1. Sample restricted to labor force participants, i.e. respondents who either held a job or were actively seeking one at the time of the survey. Respondents were sampled 60% at home (one per household) and 40% at transportation hubs, alternating between male and female respondents. is defined as having access to a health insurance scheme, a pension or a written employment contract. The safety scale ranges from 1 (very safe) to 7 (very unsafe). Weekly income and revenue are reported in Nigerian Naira and weekly revenue is also reported for those who report being self-employed.

When it comes to commuting and safety, the main focus of the primary data sample, the patterns in Table 3 reveals transportation challenges faced by Nigerian urban workers in some of the largest cities of the country. The table only includes commuters or people who report not working from the same location as their home. The average commute distance is 10.4 kilometers, taking about 41 minutes and costing approximately 1000 Naira per day. However, these averages mask considerable heterogeneity as some workers may walk to relatively nearby jobs, while others undertake long commutes via public transportation or their own cars and bikes: by comparison, respondents who report walking to work spend 34 minutes commuting

an average of 4.9 kilometers per day, while all other respondents who do not commute by walking spend an average of 43 minutes commuting 11.3 kilometers.

The most common primary commuting modes are: walking (14% of respondents), buses (19%), motorcycle taxis (14%), taxis (7%), and keke napep (motorized tricycles, which are common across Sub-Saharan African major cities) (15%). Minibuses account for 10%, while private vehicles and motorcycles together account for about 18% of commuters. Less than 1% rely on ride sharing. Only 24% of respondents who commute own any transportation asset, with 11% owning a car, 11% owning a motorcycle, and 1% owning a bicycle. This low vehicle ownership rate underscores the dependence on walking and public transportation. Safety perceptions during commuting are moderate: on a scale from 1 (very safe) to 7 (very unsafe), respondents average 2.41, suggesting most feel reasonably safe but not completely secure. About 19% report having been victims of mugging, but only 11% report sexual harassment and 9% report sexual assault. Notably, these figures are similar for men and women, contrary to what might be expected if safety concerns disproportionately affect women in this context.

Table 3: Descriptive statistics of commuting by gender: only commuters

	Male	Female	All	<i>p</i>
Respondent owns a transportation asset	0.34	0.13	0.24	0.000***
Respondent owns a car	0.12	0.09	0.11	0.014**
Respondent owns a motorcycle	0.18	0.04	0.11	0.000***
Respondent owns a bicycle	0.03	0.00	0.01	0.000***
Commute distance (in kms)	10.44	10.32	10.38	0.113
Commute time in mins	41.43	40.94	41.20	0.168
Commute cost (in Naira)	996.12	1002.93	999.25	0.301
<i>Primary commuting mode</i>				
Buses	18.83	18.52	18.69	0.444
Keke Napep	13.53	17.16	15.22	0.132
Minibuses	7.58	12.35	9.80	0.004***
Motorcycle taxi	12.23	16.91	14.42	0.029**
Other	2.06	3.09	2.54	0.253
Private bicycle	1.52	0.62	1.10	0.055*
Private motorcycle	16.02	2.35	9.63	0.000***
Private vehicle/car	8.23	5.68	7.04	0.017**
Ride hailing	0.54	0.62	0.58	0.913
Taxi (car)	5.41	9.63	7.38	0.003***
Walking	14.07	13.09	13.61	0.277
Feels safe during commute (1 to 7)	2.40	2.42	2.41	0.142
Self: ever victim of mugging	0.18	0.21	0.19	0.392
Self: ever victim of sexual harassment	0.09	0.12	0.11	0.002***
Self: ever victim of sexual assault	0.09	0.09	0.09	0.806
Own gender: ever victim of theft	0.42	0.42	0.42	0.581
Own gender: ever victim of sexual harassment	0.19	0.23	0.21	0.008***
Own gender: ever victim of sexual assault	0.16	0.18	0.17	0.465
Experienced non-fatal traffic accident past 3 mo	0.07	0.04	0.06	0.002***
Acquaintance in a fatal accident past 12 mo	0.09	0.08	0.09	0.403
Weekly fuel cost (in Naira)	12567.25	14204.29	12991.67	0.019**

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column 1 and 2 report the sample means for male and female respondents respectively, column 3 reports the combined sample means and column 4 calculates the p-value for the t-test of the difference in means column 2 and column 1. Sample restricted to labor force participants and respondents who report not working from their home location. Respondents were sampled 60% at home (one per household) and 40% at transportation hubs, alternating between male and female respondents. The safety scale ranges from 1 (very safe) to 7 (very unsafe). Weekly fuel cost is reported in Nigerian Naira. Commute distance, time, cost and fuel cost are winsorized at the 1% level.

4 Empirical strategy

The main challenge is that city shape is endogenous: cities with better performing labor markets and thus better economic opportunities may attract more infrastructure investment and more compact development, or unobserved factors such as governance quality, or specific or attractive geographic features may affect both city shape and labor market performance. I address this endogeneity through an instrumental variables approach, constructing a footprint of cities only containing potentially developable pixels to isolate exogenous variation in city shape.

There are multiple potential sources of endogeneity that threaten a causal interpretation of the relationship between city shape and labor market outcomes. Colonial urban planning decisions and historical institutions may have influenced both city shapes and subsequent economic development. Baruah, Henderson, and Peng (2021) show that “colonial legacies of sprawl” continue to affect child health outcomes in African cities today, suggesting there could be persistent effects of historical spatial structure. I show in later tables robustness to considering the colonial origins of cities. Secondly, as cities grow, shapes become more and more distorted but rapid population growth in certain cities might be due to greater existing labor market opportunities, leading to reverse causality¹². Harari (2020) documents that population growth and city shape are negatively correlated across Indian cities. I include city size in the specifications to address this specific concern. Third, strong governments can promote both infrastructure and labor market policies that improve employment outcomes, confounding the relationship between city shape and employment. Finally, natural constraints and specific geographic features might both determine the shape of a city, as well as its labor market, for example if cities have beautiful sea fronts and develop a tourism industry because of it. Taken together, these concerns motivate the use of an instrumental variables approach that isolates variation in city shape driven by exogenous geographic constraints interacted with pre-determined population growth patterns.

4.1 Instrumental variable construction

Following Harari (2020), I construct an instrumental variable for city shape based on the potential developable footprint of each city. The key insight is that cities constrained by geographic barriers (water, steep terrain) will develop more distorted shapes as they grow, while unconstrained cities could expand in a more circular pattern. By using geographic barriers and projected population growth based on pre-1975 trends predating most of the

12. Although the city size wage premium has been established in US-based research (Baum-Snow and Pavan 2012; Martellini 2019) with workers in larger cities earning higher wages, the evidence from African cities shows limited evidence of an urban wage premium as in Jedwab (2013) in the first place. Recent papers do find evidence of a city size wage premium in Latin America (Barufi, Haddad, and Nijkamp 2023) or of a greater urban wage premium for women compared to men in Brazilian municipalities as in Almeida, Araújo, and Gonçalves (2022).

rapid urban growth for Nigeria, the instrument isolates variation in shape that is plausibly exogenous to current labor market conditions.

Actual city boundaries in 1975 and a linear population model can be used to predict the shape of the city in any given year after 1975. I draw the smallest bounding circle around the 1975 shape for each city c and the largest contiguous set of developable land pixels within that pixel identifies the city’s potential city shape for 1975. Here, I define a developable pixel as any pixel with no water or wetland and a slope lower than fifteen degrees. Each year, the potential shape can be derived from the contiguous set of developable pixels within the increasing bounding circles. The circles’ radii come from projected population and area based on historical population growth. The projected population is determined as follows: for each city c , I estimate a log-linear projection of population based on yearly population data from 1950 to 1975. This yields predicted population $\widehat{pop}_{c,t}$ for subsequent years t (including 2010 and 2025). Because this projected population is based solely on pre-1975 trends, it should not be affected by subsequent economic developments, infrastructure investments, or policy changes that might be correlated with both city shape and labor market outcomes. I predict the area of each city for each year, regressing the logged area for a city c in year t on population density in 1975, on linearized projected population $\widehat{pop}_{c,t}$ and year fixed effects. Specifically, I estimate:

$$\log(area_{c,t}) = \alpha \log(\widehat{pop}_{c,t}) + \beta \log(pop_{c,1975}/area_{c,1975}) + \gamma_t + \epsilon_{c,t} \quad (6)$$

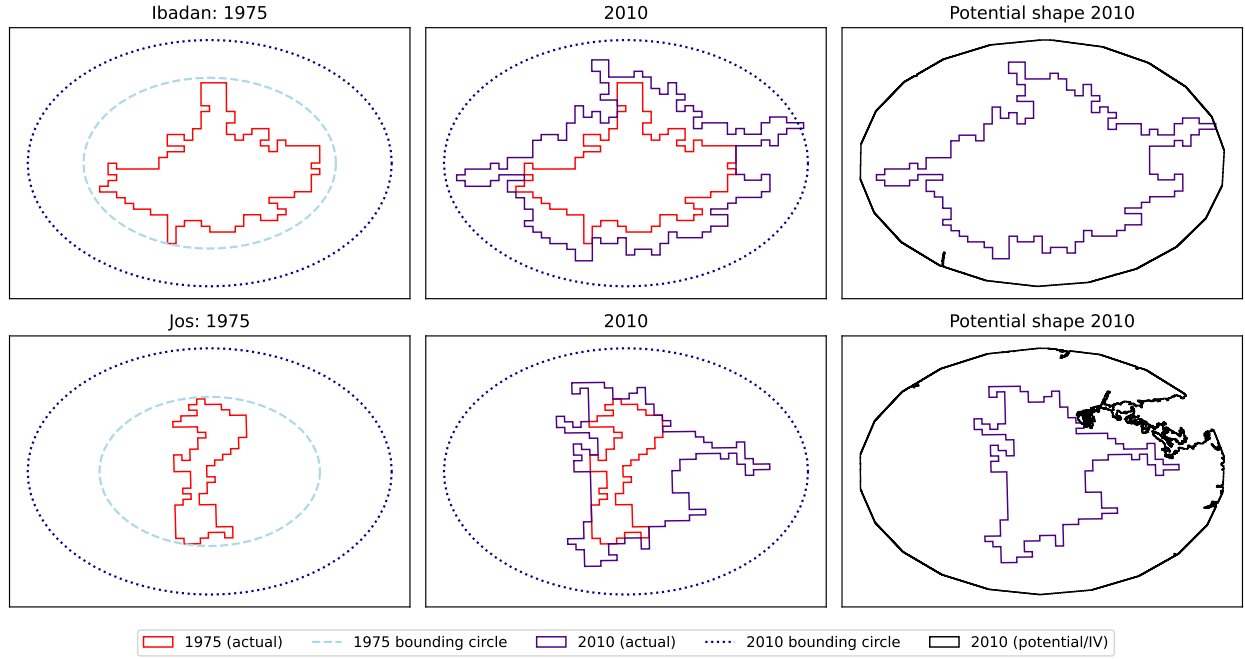
The fitted values from this regression provide predicted city area $\widehat{area}_{c,t}$. This approach assumes that cities expand their area in response to population growth, with the rate of spatial expansion depending on initial density and time trends. Finally, the predicted radius that dictates the expanding bounding circles is given by:

$$r_{c,t} = \sqrt{\widehat{area}_{c,t}/\pi} \quad (7)$$

As the predicted population (and thus predicted area) grows over time, the bounding circle defined by (7) expands outward from the 1975 city center. The potential footprint resulting from the growing circle then represents how the city would have expanded if it had kept growing at the same pace and density as before 1975 and only faced geographic constraints, without any influence from urban planning, infrastructure investments, economic shocks or governments. I then compute the same four metrics (cohesion, proximity, spin and dispersion) for this potential shape polygon and these predicted shape metrics serve as instrumental variables for the actual city shape metrics. The potential footprint approach yields a different predicted shape for each city-year based on that city’s 1975 geography and projected population growth, providing cross-sectional variation in the instrument. Intuitively, cities that are surrounded by mountains or water or wetlands, all geographic features that raise the cost of providing infrastructure, will have more distorted and less compact potential shapes

(and lower normalized shape metrics) because developable land will be fragmented due to these geographic features, whereas cities without these natural constraints would have more circular potential shapes as they can expand in all directions. This variation borne from natural constraints provides the source of identifying variation in the instrument.

Figure 3: Actual versus predicted shapes in 1975 and in 2010: Jos and Ibadan



The instrumental variables approach requires both relevance and exclusion restriction to identify a causal relationship. The relevance assumption requires that cities constrained by their 1975 geography have limited options for how they can grow, and thus the 1975 potential shape must predict the current shape. This can be directly tested via the first-stage F statistic. The exclusion restriction requires that the 1975 potential shape affects current labor market outcomes only through its effect on actual shape, and not directly. This assumption would be violated if proximity to water or mountain range affected labor markets directly and not through city shape for example. The exclusion restriction is more plausible if the potential footprint captures variation in shape that is orthogonal to other relevant geographic characteristics. I address this concern in several ways. First, I control for actual logged city area in all specifications, which absorbs some of the variation in geography that might directly affect labor markets (e.g., larger cities may have different economic structures). Second, controlling for log city area, I find that none of nine geographic characteristics, including distance to coast, ruggedness, bedrock depth, and crop suitability, are statistically associated with any of the four shape metrics, suggesting that city shape in our sample is not determined by underlying geography (see Appendix Table A3). This confirms that the

instrument uses variation in how geographic constraints shape potential expansion patterns, not just the presence or absence of constraints. Third, in robustness checks, I include controls for industry composition to account for the possibility that geographic features attract certain industries.

4.2 Specification

The main specification estimates the effect of city shape on individual labor market outcomes using two-stage least squares (2SLS). The first-stage regression is shown below for the one-round specification using the LSMS sample:

$$shape_c = \alpha_0 + \alpha_1 \widetilde{shape}_c + \alpha_2 \log(\widehat{pop}_c) + \alpha_3 X_i + \gamma_{2023} + \nu_c$$

$shape_c$ is one of the four shape metrics (cohesion, proximity, dispersion, or spin) or the standardized *Summary* index for city c in 2025. $\widetilde{shape}_c$ is the corresponding metric for the potential footprint i.e. the instrument, and $\log(\widehat{pop}_c)$ is the linearized projected population for 2025. γ_{2023} is the dummy for the year 2023. The projected population serves as an additional instrument for city area, which is included as a control in the second stage to isolate the effect of shape, holding size constant. For the primary data sample, which only includes one year of data, I exclude the year dummy.

$$\log(area_{c,2025}) = \sigma_0 + \sigma_1 \log(\widehat{pop}_c) + \sigma_3 X_i + \gamma_{2023} + \eta_c$$

The second-stage regression is:

$$Y_{i,c} = \beta_0 + \beta_1 \widehat{shape}_{c,2025} + \beta_3 \log(\widehat{area}_{c,2025}) + \beta_2 X_i + \gamma_{2023} + u_{i,c}$$

$Y_{i,c}$ is a labor market outcome for individual i in city c . The main labor market variables are the following: an indicator for being active on the labor market as defined by being employed or actively searching for a job in the past 7 days preceding the survey, employment status specifically being wage-employed, self-employed or unemployed, hours worked in the past week on the main job, and an indicator for formal employment (only defined for 2023) proxied by having access either to employer-sponsored health insurance scheme, a pension, or a written employment contract. $\widehat{shape}_{c,2025}$ is the instrumented city shape predicted from the first stage, and X_i is a vector of individual-level controls including age, years of education, and gender as well as the log of linearized projected population. Standard errors are clustered at the city level to account for correlation in outcomes within local government areas (standard errors are clustered at the LGA level for the primary data sample). The coefficient of interest is β_1 , which captures the causal effect of city shape on labor market outcomes under the identification assumptions. A positive coefficient indicates that worse shapes (higher metric values, when not normalized) are associated with higher values of the outcome variable. I

also estimate reduced-form regressions that directly relate outcomes to the potential footprint instrument:

$$Y_{i,c} = \gamma_0 + \gamma_1 \widetilde{shape}_c + \gamma_2 \log(\widehat{pop}_c) + \gamma_3 X_i + \eta_{i,c}$$

These reduced-form estimates do not rely on the strength of the first stage and provide direct evidence on whether the geographic constraints that shape potential city footprints are associated with labor market outcomes. Reduced-form estimates can be interpreted as “intent-to-treat” effects: they show the impact of being in a city with geographic constraints that would predict a more distorted shape, regardless of whether the city actually developed that shape. These estimates are informative even when the first stage is weak, making them particularly valuable in the primary data sample.

5 Results

5.1 OLS results: labor and commuting outcomes

The following tables present OLS estimates directly relating labor market outcomes to the city shape measures. Each row shows results for a different shape metric: cohesion, proximity, dispersion, spin and a summary metric (standardized to be mean 0 and standard deviation of 1). Each column shows results for the different labor outcomes: active on the labor market, self-employment, wage employment, unemployment, hours worked in the past week, formal employment and whether work is the same location as home.

Bypassing concerns about endogeneity described above, the OLS results show that worse shapes, i.e. higher shape metric values are associated with statistically significant increases in wage employment, and number of hours worked at the 10% and 1% levels respectively. Reassuringly, the association is confirmed across the different shape metrics. To interpret the magnitude of these relationships more precisely, I use the summary index measure.

Table 4 shows that a one standard deviation increase in city shape (moving towards to more distorted shapes) is associated with a 14 percentage point increase in the likelihood of being wage-employed (significant at the 10% level) and an increase of 21 hours worked per week, which is statistically significant at the 1% level. These effects are large but they also capture the fact that moving 1 SD towards worse shapes is a substantial shift in the distribution (as the standard deviation of the summary index itself is 0.95 in 2010 and 1.1 in 2023), akin to moving from Umuahia (a state capital) to Abuja (the capital) in Figure 2. I exclude labor market participation as an outcome as 85% of the respondents were already active in the sample. The OLS results however show no relationship between city shape and other features of work: no shift in self-employment or formal employment, or no change in unemployment, or where the job is located. The other associations however are somewhat consistent in sign with the prediction that worse shapes (which imply worse transportation infrastructure) would lead people to rely more on self-employment, have less access to formal jobs and

would lead them to work more hours, potentially to compensate for higher commuting costs.

Table 4: OLS: Labor outcomes, primary data sample

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage emp.	Unemp.	Nb. Hours	Formal	Same loc.
Proximity	0.017 (0.027)	0.034** (0.015)	0.001 (0.008)	4.648*** (1.414)	-0.013 (0.015)	-0.015 (0.012)
Cohesion	0.012 (0.018)	0.022** (0.010)	0.001 (0.005)	3.070*** (0.947)	-0.009 (0.010)	-0.010 (0.008)
Spin	0.000 (0.000)	0.001* (0.000)	0.000 (0.000)	0.087*** (0.029)	-0.000 (0.000)	-0.000 (0.000)
Dispersion	0.011 (0.017)	0.018* (0.010)	0.001 (0.005)	2.796*** (0.931)	-0.010 (0.010)	-0.010 (0.007)
<i>Summary</i>	0.088 (0.122)	0.140* (0.072)	0.008 (0.036)	21.054*** (6.668)	-0.070 (0.069)	-0.069 (0.055)
Observations	2462	2462	2462	2462	2462	2462

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate OLS regression of the outcome (column) on the shape (row) and includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population using the primary data sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes.

The OLS results using the LSMS sample differ from the primary data sample (Table 5): moving towards worse shapes is associated with a statistically significant and substantial decrease in hours worked by 17, which is the opposite sign of the results in the 13-city sample. Furthermore, there is no conclusive evidence on changes in labor market participation based on city shape in this sample. I therefore present robustness checks without city fixed effects, identifying off cross-city variation (see Appendix tables for reference). Since manufacturing share and colonial origin status are city-level characteristics, city fixed effects would absorb their variation. As a result, coefficients differ in sign and magnitude from the preferred specification, reflecting the difference between cross-sectional and within-city identification: cities with worse shapes tend to be larger and more economically developed, generating positive cross-sectional associations with hours and formal employment. The within-city estimates in the preferred specification isolate the effect of changes in shape, holding fixed time-invariant city characteristics.

I also use the commuting module to examine how city shape correlates to broad transportation patterns. Table 6 presents OLS estimates relating the commuting variables to city shape metrics. The outcomes include commute time (in minutes), commute distance (in kilometers), commute costs (in Naira per week), all three winsorized at the top 1%, safety during commuting (on a scale of 1 to 7, where 1 is very safe and 7 is not safe at all), as well as an

Table 5: OLS: Labor outcomes, LSMS sample

	(1) Active	(2) Self-emp.	(3) Unemp.	(4) Nb. Hours	(5) Formal
Cohesion	-0.052 (0.040)	-0.005 (0.029)	0.010 (0.008)	-3.178*** (1.042)	0.002 (0.002)
Proximity	-0.052 (0.053)	-0.013 (0.037)	0.011 (0.009)	-3.685*** (1.224)	0.003 (0.002)
Spin	0.000 (0.001)	-0.001 (0.001)	0.000 (0.000)	-0.064* (0.035)	0.000 (0.000)
Dispersion	-0.009 (0.030)	-0.011 (0.023)	0.007 (0.005)	-1.844** (0.885)	0.002 (0.002)
<i>Summary</i>	-0.129 (0.263)	-0.110 (0.185)	0.061 (0.044)	-17.053** (7.310)	0.015 (0.011)
Observations	7136	7136	7136	5983	3754

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Each cell reports the coefficient from a separate OLS regression of the outcome (column) on the shape (row) and includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population and a year dummy using the LSMS sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values of the metrics and *Summary* indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract, and is only collected in 2023 LSMS.

indicator for walking as the primary commuting mode and an indicator for using public transportation as the primary mode. Public transportation here is defined as included taking the bus or the minibus for commuting to work. Here I follow usual definitions and exclude privatized or semi-privatized modes of transportation, such as three-wheeled vehicles and motorcycle taxis that serve as non-household owned private modes of transportation but are common modes of transit for urban dwellers in Nigeria that I define below as “informal transportation”.

The results show that worse city shapes are significantly associated with higher self-reported commuting costs, as well as greater use of public buses and minibuses, and with reduced incidence of walking to work. This is consistent with the idea that worse shapes create worse accessibility and connectivity, pushing people towards more expensive modes of transportation and away from walking. Focusing on the summary index, the results suggest that a one standard deviation shift towards a worse shape is associated with a 44 percentage point increase in public transportation and a 17 percentage point decrease in informal transportation, relative to a “baseline” of 27% of respondents taking buses or minibuses as primary mode of transportation and 29% of respondents using informal transportation. Conversely, there is a 9 percentage point decrease in walking, which makes sense as more distorted shapes yield cities where distances might be longer and where walking becomes more onerous or where more people start working from home rather than commute to a job. The coefficient in column (4) for cost also implies that 1 SD increase in city shape distortion is associated with an additional 454 Naira spent on commuting (equating to an additional 2% of reported weekly average income spent), from average commuting costs of 854 Naira across the whole sample (in Table 3).

This shows that even with relatively modest to inexistent labor market effects, city shape does seem to matter for travel patterns within the cities. Interestingly, in this sample, the results show no incidence of city shape on time or distance or safety, although the coefficients are positive and thus in line with the hypothesis. Additionally, if workers in cities with worse shape adjust by choosing to work closer to home or relocate towards employment centers, the observed commuting distances and time might not differ much across cities despite large variation in city shape. The lack of finding on safety is consistent with the fact that men and women report similar levels of safety and suggest that other factors other than gender or city shape might determine feelings and perception around safety in this context. The larger coefficients and pattern might be driven in part by Lagos, as it is one of the cities with a developed transit infrastructure (the only Nigerian city with rail infrastructure as of 2024) so I exclude all Lagos observations in Table A4 and show that worse shapes are associated with less walking (-30 p.p., significant at the 10% level). As Nigeria’s largest city, Lagos might be driving some of the muted results on safety and lessening the influence of city shape on walking. Looking at the sign of the coefficient, the pattern of higher costs and more public transportation/less informal transportation use in worse-shaped cities seems robust to

Table 6: OLS: Commuting outcomes, primary data sample

	(1) Time	(2) Distance	(3) Safety	(4) Cost	(5) Walk	(6) Public tr.	(7) Informal tr.
Proximity	2.125 (2.716)	0.349 (0.526)	0.037 (0.056)	97.822*** (25.023)	-0.021** (0.009)	0.093*** (0.013)	-0.034** (0.014)
Cohesion	1.448 (1.802)	0.253 (0.355)	0.024 (0.038)	66.233*** (16.626)	-0.014** (0.006)	0.063*** (0.008)	-0.023** (0.009)
Spin	0.033 (0.048)	0.006 (0.009)	0.001 (0.001)	1.911*** (0.484)	-0.000** (0.000)	0.002*** (0.000)	-0.001** (0.000)
Dispersion	1.002 (1.514)	0.190 (0.320)	0.022 (0.036)	60.783*** (16.133)	-0.012** (0.006)	0.061*** (0.007)	-0.024*** (0.008)
<i>Summary</i>	8.683 (11.866)	1.526 (2.338)	0.162 (0.262)	454.050*** (115.968)	-0.093** (0.043)	0.441*** (0.056)	-0.166** (0.064)
Observations	2324	2434	2462	1833	1734	1734	1734

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate OLS regression of the outcome (column) on the shape (row) and includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population using the primary data sample. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values in the shape metrics indicate worse city shapes. The safety variable is a Likert scale of respondents reporting how safe they felt during their commute, from a scale from 1 to 7, with 7 being the least safe. The commuting time, distance and cost variables are winsorized at the top 1% percent. Distance is expressed in kilometers, time is expressed in minutes and costs are expressed in Nigerian Naira. Public transportation is defined as commuting by bus or minibuss; informal transportation is defined as commuting by motorcycle or tricycle taxis.

excluding Lagos (although no longer significant), although much more imprecisely estimated.

The opposite signs in OLS for the two different samples suggest that they might be subject to different types of selection based on the sample of cities they are composed of. I look at city-level regressions that albeit biased are more informative on the between-city variation that city shape aims to capture, rather than the within-city variation. Full table for the *Summary* is in Appendix (Table A5), showing that between-city variation drive most of the coefficient on hours, employment type, work location, industrial mix and firm size to some extent but shows no clear pattern for commuting in the primary data sample; in the LSMS sample, self-employment and firm size as well as industrial mix seem to matter for cross-city variation.

5.2 IV results: labor outcomes

5.2.1 First-stage results

The tables below present first-stage estimates relating actual city shape metrics to the potential footprint instrument. Each column shows results for a different shape metric: proximity, spin, cohesion, and dispersion. I regress the potential shape metrics, derived from computing the shape metrics on the potential footprint derived from projected population growth and the geographic constraints the city encounters as it is growing. The regression also includes correlates for age, gender and education and logged area. In both samples, the potential footprint is a strong predictor of actual city shape. F-statistics range from 55 to 199 across all metrics in the primary data sample, well above the conventional threshold of 10 for strong instruments in Tables A8. This holds even when excluding controls (see Table A16). The coefficients on the potential footprint are positive and significant for all metrics. This suggests that cities constrained by geographic barriers in 1975 indeed developed more distorted shapes as they grew.

5.2.2 IV results

Table 7 presents two-stage least squares estimates of the effect of city shape on labor market outcomes in the primary data sample, including respondent controls and logged area. The table also reports the first-stage F-statistic for each specification. The standard errors are clustered at the LGA level. The IV results are mostly insignificant across all shape metrics and outcomes. Point estimates are imprecise, and confidence intervals are wide, despite the strong first-stage relationship discussed above. The wide standard errors reflect the limited power of the IV approach in this smaller sample. Key exceptions here are formal employment or the probability of the respondent working from home, which are both statistically significant at the 10% level for the *Summary* metric. Examining the sign of some of the

coefficients despite the lack of significance, worse shapes would cause greater prevalence of self-employment and an increase in the number of hours worked, while lowering the likelihood of wage employment. As before, the large magnitudes reflect both the range of the distribution of city shape (moving one standard deviation represents a large shift), as well as imprecision in the second stage of the IV due to the small number of clusters in the primary data sample.

Table 7: IV: Labor outcomes, primary data sample

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage empl	Unemp.	Nb. Hours	Formal	Same loc.
Proximity	0.180 (0.143)	-0.100 (0.112)	0.008 (0.015)	1.671 (7.237)	-0.116 (0.082)	-0.070 (0.049)
Cohesion	0.117 (0.089)	-0.063 (0.070)	0.006 (0.010)	1.178 (4.744)	-0.076 (0.051)	-0.046 (0.031)
Spin	0.002*** (0.001)	-0.000 (0.000)	0.000 (0.000)	0.061 (0.059)	-0.001*** (0.000)	-0.001* (0.000)
Dispersion	0.219 (0.378)	-0.186 (0.344)	0.018 (0.022)	-2.210 (10.588)	-0.132 (0.217)	-0.081 (0.117)
<i>Summary</i>	0.682 (0.431)	-0.368 (0.353)	0.030 (0.057)	8.863 (25.910)	-0.417* (0.237)	-0.264* (0.158)
Dep. var. mean (%)	42.9	24.9	5.8	36.0	20.4	29.2
<i>Obs.</i>	2462	2462	2462	2462	2462	2462
<i>F</i> -statistic	114	114	114	114	114	114

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape, using the primary data sample. The F-stat for the 2SLS specification using the *Summary* metric is shown in the bottom row for the corresponding outcome. The specification also includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Sample includes all respondents active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. “Same location” is defined as reporting working from the same address as one’s home location.

Turning to the LSMS sample does not provide much more clarity despite the much larger number of cities. The labor market outcomes here are restricted to being active on the labor market, self-employment and unemployment, hours worked, and working in a formal job (as defined by having a written contract or health insurance scheme through their employer or a pension). Worse shapes tend to move people away from the labor market and from self-employment and unemployment. Using the summary measure, a one standard deviation increase away from mean city shape leads to a 17 percentage point decrease in probability of being self-employed (significant at the 10% level), from a baseline of 58% in 2010 and a 3 percentage point decline in unemployment (not statistically significant), which would be consistent with worse shapes moving people away from the labor markets. There is

suggestive evidence that worse shapes also lead to exit from the labor market, with a 1 standard deviation increase in the summary measure leading to a 8 percentage point decline in labor market participation, i.e. either being employed or unemployed and actively searching for a job, along with suggestive evidence that worse shapes lead to 4 more hours worked, suggesting that workers who stay in the labor market in worse-shaped cities might work more hours in order to compensate for higher transportation costs. The confidence intervals are too large to decisively rule out any clear effect of city shape on labor outcomes overall.

Table 8: IV: Labor outcomes, LSMS sample

	(1) Active	(2) Self-emp.	(3) Unemp.	(4) Nb. Hours	(5) Formal
Cohesion	-0.014 (0.010)	-0.028* (0.015)	-0.005 (0.004)	0.651 (0.521)	0.001 (0.004)
Proximity	-0.021 (0.015)	-0.042* (0.022)	-0.008 (0.006)	0.965 (0.780)	0.002 (0.006)
Spin	-0.000 (0.000)	-0.001** (0.000)	-0.000 (0.000)	0.016* (0.009)	0.000 (0.000)
Dispersion	-0.022 (0.019)	-0.041 (0.029)	-0.009 (0.007)	0.827 (0.796)	0.000 (0.005)
<i>Summary</i>	-0.082 (0.063)	-0.173* (0.091)	-0.031 (0.023)	4.258 (3.193)	0.009 (0.026)
Dep. var. mean (%)	63.3	30.1	4.3	28.9	9.4
<i>Obs.</i>	7136	7136	7136	5983	3754
<i>F</i> -statistic	106.95	106.95	106.95	96.51	74.19

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape using the LSMS sample. The F-stat for the 2SLS specification using the *Summary* metric is shown in the bottom row for the corresponding outcome. The specification also includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population and a year dummy. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract and is only collected for 2023 LSMS. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Overall results suggest limited incidence of shape on labor market outcomes, with the slight changes being restricted to the intensive margin (move away from wage/self-employment, formal jobs) rather than on the extensive margin. I also look at other work outcomes such as wages, revenues and sectoral shares (manufacturing, services and low-productivity services) reconstructed from LSMS data at the city level and also find no incidence of city shape (see Table A9). I examine robustness to colonial origin using (Baruah, Henderson, and Peng 2021)’s dataset and colonial origins variable: only 3 cities in either the LSMS sample and the

13-city sample (Enugu, Abuja and Ibadan) are flagged as having colonial origins. Despite limited statistical power, I show in the appendix that IV results are virtually unchanged when excluding these cities. Results are robust to excluding controls, although excluding controls leads to a weaker first stage (Table A16), or to controlling for industrial mix at the city level (see robustness table: Table A18 for LSMS and for primary data sample, Table A17).

Finding limited incidence that city shape affects worker behavior, I look for evidence that firms rather than workers adjust to city shape. I use the publicly available 2023 Nigerian Business Census data and compute average number of employees per firm for all the firms in the census. There is some evidence of a correlation between firm size and shape metrics (see Figure A5) and city-level IV using the LSMS sample (Table A10) shows a hollowing out of the firm size distribution in worse-shaped cities, with 1 SD increase in distortion increasing the city-level share of 1-person firms by 18 percentage points at the 5% confidence level, and the share of large firms (with more than 100 employees) by 0.5 percentage points, statistically significant at the 5% level. Although not statistically significant, the coefficient on the share of micro (1-5 employees) firms is large and positive (+5 percentage points), while the coefficients on small (5-20 employees) and medium (21-99) firms are negative and around -2.8 percentage points for each. These results are sensitive to the inclusion of city-level controls for age, gender and education: without controls (Table A19), the signs on 1-person and micro firms reverse and the coefficients on 1-person firms lose significance, although the effect on large firms remains robust and there is now a statistically significant increase in the proportion of small firms (5-19 employees). The preferred specification is the one with controls as it helps account for differences in composition across different cities that might be correlated with city shape. This finding on firm size can be explained by the self-employment patterns as shape worsens, making it a more difficult endeavor in worse-shaped cities perhaps by complicating access to vendors, suppliers and customers for small and medium-sized firms, but moving people towards self-employment from home without any employees. One way to read this result is that worse accessibility requires both higher transportation costs and more difficult logistics, which larger firms would have an easier time incurring, and pushes 1-person firms closer to home, and as such, the results partially reflect selection in which type of firms can survive in worse-shaped cities.

These null IV results could reflect either a true absence of effects or lack of statistical power to detect modest effects due to the smaller sample, even as the OLS results provide suggestive evidence of positive (but not meaningful) associations between worse shapes and some of the discussed labor market outcomes. As exploration, I run a simulation to detect the MDE in my research design and find that the 13-city sample only have 25% power, with a MDE that is 2.9 times the IV point estimate. For these reasons, I mostly focus on the LSMS sample moving forward and show the equivalent primary data sample tables in appendix.

Furthermore, examining the difference in sign across OLS and IV models provide some information about the type of selection at play in both samples. In the LSMS sample, OLS and IV results are of the same sign, suggesting attenuation bias in OLS with OLS models understating the negative effect of city shape on labor markets due to OVB or unobserved characteristics that help households and individuals compensating for the negative effect of shape. These characteristics might be related to housing markets or the specificity of the informal and formal labor markets in the different cities, with workers having access to better work conditions even while informal or self-employed if for example they are able to work from more organized marketplaces and have access to a larger customer base. This suggests that city shape might still affect labor markets, not by switching workers across different types of employment (wage employment vs self-employment, informal vs formal) but by changing their work configurations within the same employment type (e.g. working from home as opposed to marketplaces). In the primary data sample, OLS and IV results are of different sign (for most of the outcomes), suggesting upward bias in OLS which is coherent with workers selecting into worse shaped cities. This makes sense given the sample of cities: larger cities have worse shapes but also most likely more labor market opportunities i.e. Ibadan, Abuja, Lagos. Differences in the sign of the coefficient in OLS and IV across samples suggest that they have a different composition, which I will explore further in the next section.

Table 9: OLS/IV comparison, LSMS sample

	Active		Self-emp.		Unemp.		Hours		Formal	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	OLS	IV	OLS	IV	OLS	IV	OLS	IV	OLS	IV
<i>Summary</i>	-0.325	-0.082	-0.435***	-0.173*	0.007	-0.031	-76.747***	4.258	-3.486***	0.009
	(0.232)	(0.063)	(0.149)	(0.091)	(0.030)	(0.023)	(12.272)	(3.193)	(0.350)	(0.026)
<i>Obs.</i>	7136	7136	7136	7136	7136	7136	5983	5983	3754	3754
<i>Durbin-Wu-Hausman test</i>										
<i>p-value</i>	0.329		0.010***		0.025**		0.002***		0.540	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors clustered at the city level. Each column pair reports OLS and 2SLS estimates from a separate specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape, using the LSMS sample. Controls include age, gender, years of education, log of linearly projected population and a year dummy. *Summary Index* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values of the index indicate worse city shapes. “Active” means active on the labor market in the past week, including both employed and job-seeking respondents. “Self-emp.” denotes self-employment status. “Unemp.” denotes unemployment status conditional on labor force participation. Hours worked are set to zero for unemployed workers and null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. The Durbin-Wu-Hausman test uses a control-function approach; rejection of the null may reflect endogeneity of the OLS estimate (omitted variable bias) or divergence between the local average treatment effect and the average treatment effect.

5.3 Heterogeneous effects: gender, migration status and centrality

5.3.1 Gender differences

One of the primary objectives of this paper is to study the role of gender in the relationship between transportation infrastructure and labor market outcomes in Nigeria’s major cities.

I find a lack of gender differences in how workers respond to city shape in this context: in the 13-city sample, results are insignificant as for the main specification shown in Table A11. There is no evidence here that women react differently to city shape than men in these thirteen Nigerian cities.

I show that in the LSMS sample, there are gender differences when shape worsens. The evidence in Table 10 suggests that women are more responsive to city shapes than men within the same cities. In cities with worse shapes, the *Female* $\times$ *Summary* coefficient indicates that worsening shapes are relatively less detrimental for women workers relatively to men, as women are more likely than to be active on the labor market (not significant), which is consistent with the finding that worse shapes increase women’s likelihood of being unemployed (statistically significant at the 5% level). The effect remains not economically significant with only an increase of less than 1 percentage point in unemployment when city shape worsens by one standard deviation. Conversely, women see worsened conditions on the labor market when shape worsens, with a statistically significant decrease in formal employment by approximately 5 percentage point at the 1% level, relative to a baseline of 6% for female respondents in LSMS 2023, respectively. Looking at gender gaps in labor income in Table A13, there are no significant differences by gender in reported wages or revenues for self-employed people.

Given the differences in city size between the two samples, it might be that neither LSMS or primary data samples are powered enough to detect gender differences that depend on city size variation, especially if gender differences might be more acute in smaller cities than in larger cities, where there is a wider array of opportunities even for women more confined to the domestic sphere. I use the LSMS data to look at this possibility, investigating the city size gradient in gender differences in labor outcomes in Figure 4. The existence of a city size gradient seems most plausible for being active on the labor market, self-employed or unemployed as the figure suggests some degree of convergence in larger cities for these outcomes.

I investigate this for the self-employment and labor market participation in this Table 11 as well as in appendix (A12) for the remaining outcomes. Splitting the sample between smaller and larger cities, above and below the 300,000 threshold (used prior to 2025 by the United Nations Population Division to determine major cities) is consistent with city size contributing to limited gender heterogeneity for labor outcomes. A 1 SD increase in city shape distortion yields a decrease in the likelihood of being self-employed for men (-18 pp in cities of less than 300,000 inhabitants and around -2 pp in larger cities, although the

Table 10: IV: Heterogeneity by gender, LSMS sample

	(1) Active	(2) Self-emp.	(3) Unemp.	(4) Nb. hours	(5) Formal
<i>Summary</i>	-0.092 (0.064)	-0.183** (0.092)	-0.034 (0.023)	4.048 (3.191)	0.035 (0.025)
Female x <i>Summary</i>	0.020 (0.014)	0.018 (0.013)	0.006** (0.003)	0.409 (0.572)	-0.051*** (0.007)
Female	-0.056*** (0.016)	0.054*** (0.018)	-0.005 (0.005)	-2.687*** (0.844)	-0.012 (0.011)
<i>N</i>	7136	7136	7136	5983	3754

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Estimates from a 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape using the LSMS sample. Controls for age and number of years of education, and log of linearly projected population are included as well as a year dummy. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values in the *Summary* index indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract, and is only collected for 2023 LSMS. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

coefficient is only statistically significant in the larger cities at the 10% level). For female respondents, however, a 1 SD increase in city shape distortion causes a slight increase in the likelihood of being self-employed, statistically significant at the 5% level in large cities. For women in small cities, the net effect of city shape on self-employment is 24 percentage points, whereas for women in large cities, it is close to zero. I however fail to reject the null that this interaction coefficient is the same in small and large cities ($p = 0.134$). The positive (but not statistically significant) effect that shows up in the main gender specification in Table 10 appears to be primarily driven by women in cities of less than 300,000 inhabitants, where they may be more constrained in their ability to compensate for worse shapes, leading them to substitute into self-employment.

5.3.2 Other outcomes: migration, neighborhood quality, safety and centrality

Beyond gender, I also examine the role of migration status, neighborhood characteristics and distance to city center in explaining response to city shape. I focus the discussion of the results on the primary data sample for the variables exclusively available in the primary data sample.

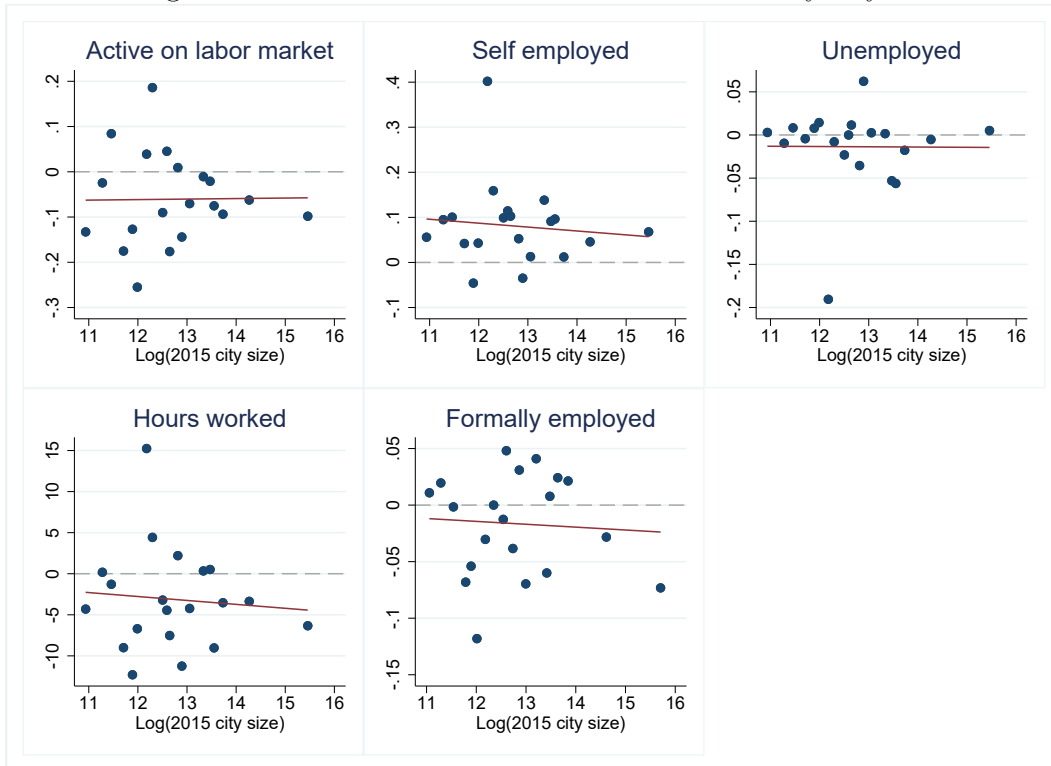
Migration status Using information and preferences collected, I find that migrants and locals seem to make different residential decisions, with migrants more likely to report choos-

Table 11: IV: Heterogeneity by gender and city size, LSMS sample

	Active		Self-employed	
	(1) Small (<300k)	(2) Large ($\geq$ 300k)	(3) Small (<300k)	(4) Large ($\geq$ 300k)
<i>Summary</i>	-0.3134 (0.2139)	-0.0300*** (0.0092)	-0.1800 (0.2300)	-0.0192* (0.0108)
Female	0.2672 (0.2318)	-0.0612*** (0.0206)	0.3541* (0.1974)	0.0574*** (0.0208)
Female $\times$ <i>Summary</i>	0.4447 (0.3072)	0.0286*** (0.0090)	0.4206 (0.2697)	0.0198** (0.0087)
Male control mean	0.6621	0.6639	0.2518	0.2749
[<i>small</i> = <i>large</i>] p-value	0.172		0.134	
<i>Obs.</i>	1932	5204	1932	5204

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Estimates from a 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape with the LSMS sample. Each column corresponds to a regression of the self-employed indicator on the cohesion or the summary metric interacted with the female dummy, separately for cities with population below and above 300,000 people. Controls for age and number of years of education are included, as well as a year dummy. City population is taken from the UN World Urbanization Prospects data and from citypopulation.de. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Figure 4: Gender differences in labor outcomes by city size



ing their home location because of affordability, proximity to jobs, rather than because of proximity to family compared to locals. Moreover, migrants seem to live in “better” neighborhoods (see detailed discussion in Appendix A.2). Despite these differences, as I show in Table A14, the coefficients show that there are clear differences between LGA natives and non-natives but few of the labor outcomes get significantly or meaningfully impacted by changes in city shape for natives. For LGA natives, worse shapes further increase the likelihood of being formal (+8 p.p. at the 5% significance level): in short, worse shapes seem to move LGA natives to move slightly closer to non-natives’ labor market behavior. I take this as evidence that residential sorting could potentially dampen the adverse effects of city shape on labor markets since I find null coefficients across the board for migrants but not for natives as city shape changes, and migrants are able to more freely choose their home location taking shape into account. This is the case despite no evidence of the fact that migrants seem to have better commutes than locals which points to potential limitations to residential sorting in this context (specifically, migrants incur even higher costs of commuting when shape worsens beyond the increase for locals as well).

Centrality vs periphery Similarly, I try to exploit within-city variation by comparing central and peripheral LGAs. I do this with the intuition that for very peripheral LGAs, and for far-flung workers, city shape might matter more than for people living next to central business districts, as city shape might heavily influence both extent and existence of transportation infrastructure on which far flung workers will depend to commute. This allows me to focus on a lower level of variation than the city-level variation that the city shape instrument is relying on.

I show the results in Table 12 for the *Summary* measure and find few distinctions between central and non-central LGAs with respect to the role of city shape in determining labor market outcomes. For LSMS respondents living in more central locations, worsening shapes lead to slightly higher lower unemployment (by around 1.5 percentage points), significant at the 5% level. Leveraging precise job locations in the primary data sample, I find that the lack of results here could be due to a concentration of jobs within 10 kilometers of Central Business Districts across most of the cities in the sample (see Figure A7). This could also be driven by the differential role of public versus informal transit, where further away workers rely on public transportation, whereas more central workers or commutes around the city center rely on informal transportation, whose use steeply declines as shape worsens (as per Table 6).

Neighborhood quality, safety and services I also try to investigate whether there is heterogeneity based on neighborhood-level characteristics that could help explain the labor results. I use the data collected to compute a standardized index of neighborhood quality based on reported presence of utility infrastructure and another index of amenities at the

Table 12: IV: Heterogeneity by centrality, LSMS sample

	(1) Active	(2) Self-emp.	(3) Unemp.	(4) Nb. Hours	(5) Formal
<i>Summary</i>	-0.011 (0.010)	0.012 (0.013)	0.001 (0.004)	2.288*** (0.725)	0.013 (0.009)
Central LGA x <i>Summary</i>	0.027 (0.022)	0.035 (0.024)	-0.015** (0.008)	-0.166 (0.974)	0.001 (0.014)
Central LGA	-0.051*** (0.017)	-0.068* (0.040)	0.018 (0.012)	-4.119*** (1.502)	0.001 (0.019)
<i>Obs.</i>	6523	6523	6523	5568	3754

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape with the LSMS sample. The specification also includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population and a year dummy. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Central LGAs are defined as LGAs that have average building height above the 75th percentile compared to other LGAs within the same city and are defined as 0 for cities with only one LGA. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values for the shape metrics indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract and is only collected for 2023 LSMS.

neighborhood level. I leverage the information on safety and create a safety index using victimization self-reports for theft, mugging, sexual harassment, sexual assault. Tables on infrastructure and safety in appendix (panels A and B in Table A15) show the results for continuous indices. Overall, there are no uniformly differential effects of city shape by neighborhood services or safety, except for some suggestive evidence on self-employment for safety and neighborhood services indices.

The results suggest that in worse shaped cities, living in a “better” neighborhood, i.e. with more services or improved safety, leads workers to exit self-employment (by 4 and 1 percentage points significant at the 1% or 5% level for safety and services respectively). Safer, better-equipped neighborhoods in Nigerian cities tend to be residential areas often located away from commercial centers and marketplaces. In this context, the interaction between safety and shape might deter self-employment because self-employment from home might not be too lucrative as customers and suppliers are far (in the center) and commuting through unsafe areas in worse cities might be undesirable. However, this hypothesis seems plausible but would require to fully disentangle the effect of safety and neighborhood location within the city, which is difficult with the data available in the LSMS. Moreover, this warrants some caution as this pattern is only found in self-employment and does not seem to replicate for the other labor outcomes.

6 Discussion and conclusion

6.1 Summary of findings

The OLS results across both samples confirm the finding from Harari (2020) and Duque et al. (2022) that worse shapes are associated with worse accessibility: workers in more distorted cities face higher commuting costs and are more likely to rely on public transportation and less on walking. On the labor market side, results are more muddled with worse shapes in the primary data sample being associated with statistically significant increases in wage employment and hours worked and in the LSMS sample with decreases in hours worked, although the results are not robust to excluding demographic controls and including controls for colonial origin or manufacturing share due to differences in specification (exclusion of city-level fixed effects). Focusing on cross-city variation, city shape is associated with a change in hours, work location and firm size in the primary sample and in self-employment and firm size as well as industrial mix in the LSMS sample.

However, limited power in my 13-city sample and issues of endogeneity limit the validity of the OLS only approach. However, limited power in my 13-city sample and issues of endogeneity limit the validity of the OLS only approach. Since endogeneity or differences between the local average treatment effect and the average treatment effect are concentrated in the outcomes where OLS and IV diverge most (Angrist and Pischke 2009; Lochner and Moretti 2011): in the LSMS sample, where OLS and IV are of the same sign, attenuation bias with OLS understates the negative effect of shape due to unobserved characteristics that may help workers compensate. In the primary sample, where OLS and IV are of opposite sign for most outcomes, there is positive selection into worse-shaped cities that also happen to be the largest and most economically dynamic.

The IV results in the primary data sample are mostly insignificant, reflecting the limited statistical power rather than necessarily a true absence of effects. Relying on the larger LSMS sample yields the finding that worse shapes affect only self-employment (at the 10% level). There is also suggestive evidence of more hours worked, suggesting that workers who stay in the labor market in worse-shaped cities might work more to compensate for higher transportation costs. The confidence intervals remain wide and do not allow to decisively rule out any clear effect of city shape on labor outcomes overall. Inclusion of city-level manufacturing share weakens the self-employment finding and yields noisier estimates in the primary data sample. Results are otherwise robust to excluding controls, with the first stage remaining strong, or to considering colonial origin.

Looking at heterogeneous treatment effects yields more informative patterns. I find some gender differences in response to city shape in the LSMS sample: women are less likely to hold formal jobs when shapes worsen. Although not statistically significant due to the small number of clusters, the evidence suggests a potential city size gradient mediating gender differences in the labor market, consistent with recent research in Latin American countries

highlighting similar patterns (Barufi, Haddad, and Nijkamp 2023). The rich data collected in the primary data sample also highlights labor market differences between LGA natives and non-natives, showing that worse city shapes push LGA natives to behave more like migrants within the same labor markets, consistent with the fact that natives’ residential locations are more likely to be exogenously determined relative to labor market considerations.

Finally, attempts to exploit within-city variation yield mixed results, whether using LGA average building height as a measure of centrality, constructed indices for neighborhood characteristics, or safety indices. There is some solid evidence that in safer, better-equipped neighborhoods, worse shapes reduce self-employment, potentially because these residential neighborhoods are located away from commercial centers making home-based work less viable and commuting through the city less attractive. Contrary to Harari (2020)’s findings on lack of adjustment by firms in India, I find here adjustment of firms to city shape, with the distribution becoming more skewed (more 1-person firms and more large firms). Across all the subsamples for gender, migration, centrality, neighborhood-level outcomes, self-employment responds the most consistently to worsening shapes in the IV models, which might point out at the differential selection into self-employment which is beyond the scope of the data available in this paper.

All in all, the limited incidence of city shape on labor outcomes, the clear effect on commuting costs, and the role of city size in mediating gender differences suggest that the main adjustment channel in this context operates through something other than extensive margin changes in labor markets, a point I return to in the discussion below.

6.2 Discussion

Taken all together, the results imply that city shape distortion seems to influence work configurations by changing the way and where people work, rather than external margin changes or switching across employment types as documented in South Asia (Harari 2020) or in Latin America (Duque et al. 2022), although the confidence intervals remain too wide to rule out effects of similar magnitude.

If shape is relevant to transportation and accessibility in these cities, it could be workers in complier cities, i.e. cities whose current geographic footprint is highly determined by the geographic obstacles (rather than colonial legacies, planning or other economic forces) cities met in expanding from their 1975 boundaries, have already adapted and for the most part, marginal changes in city shape would not further affect their behavior on the labor market.

This idea that workers may have fully absorbed the cost of city shape is consistent with my finding that these cities see a large proportion of workers engaged in work from home and/or in self-employment. The firm size results reinforce the interpretation that shape operates through spatial reorganization rather than employment switching: worse shapes do not

eliminate economic activity but hollow out the middle of the firm size distribution, simultaneously pushing workers toward sole operation from home and favoring large firms with the scale to absorb higher logistics costs, while squeezing out the small and medium firms in between. As worse shapes may increase the cost of commuting, workers can adapt by staying close to home or traveling to the most accessible location such as CBDs¹³. Looking at the concentration of jobs from the primary data sample seems to confirm that workers absorb the higher cost of commuting of these cities, with close to 80% of employment being located within 10 kilometers of the city center in most cities. Despite the limited power in the 13-city sample, relying only on cross-city variation does not eliminate the positive effect of shape on likelihood of working from home, suggesting potential adjustment through residential sorting.

Furthermore, since workers do not respond on the labor market or via their transportation choices (for the most part), housing and residential choice matter in this context, with workers being attached to job locations and moving housing, rather than the reverse. The lack of significance on commute distance and time in the IV and in OLS point to a pattern where workers adjust their residential location and thus keep their commutes manageable both in terms of time and distance but in worse cities, different modes of transportation may increase the cost of one unit of commuting distance leading to increased commuting costs for worse-shaped cities. This residential optimization is however constrained by multiple factors: availability of housing across different types of locations (48% of the primary data sample lives in informal settlements by their own reporting), affordability constraints that are well documented in the literature for Sub-Saharan Africa (Parby et al. 2015), and individual tastes (family ties and social networks determining their home location, also documented in Rojas Ampuero and Carrera (2025) and Barnhardt, Field, and Pande (2017)). With perfect sorting, there should be no incidence of shape on any of the commuting outcomes. These constraints also show up in the form of the higher costs faced by migrants when commuting comparatively to natives, the fact that worse shapes do seem to be associated with higher commuting costs as well as the negative effect of the interaction between safety and shape on self-employment, where “better” neighborhoods might be chosen at the expense of other considerations such as access to commercial centers and workplaces or to customers and suppliers.

Finally, the results also reveal the contribution of city size in mediating the effects of shape, at least for women. These results are worth highlighting in a context where in Sub-Saharan Africa and in Central and Southern Asia, small and medium-sized cities (fewer than 250,000 people) and fewer than 1 million people, respectively) are experiencing the fastest growth in recent years, bucking urbanization trends from the past decades in Asia and Latin

13. I also find very limited variation in commuting distance by number of years lived in the city or by quartile of city shape, further confirming that workers have already responded to bad shapes by commuting as far as they can (not shown here).

America, where “megacities” were primary drivers of urban growth.

As these cities continue to grow, attention has to be paid to their spatial organization as population has been outpaced by growth in built-up area United Nations, Population Division (2025) and as small cities in West Africa have been expanding with more sprawl compared to large cities since 2001 according to Korah and Wimberly (2024). The contribution of urbanization to economic growth is well-known (Bryan, Glaeser, and Tsivanidis 2020) and despite some evidence of stalled productivity growth in African cities compared to their counterparts in other continents (Gollin, Jedwab, and Vollrath 2016), urbanization has led to 30% of the increase in GDP per capita since 2001 in African cities (OECD/UN ECA/AfDB 2022). Given the outsized role small- and medium-sized cities will play in urbanization and urban growth, there needs to be urgent policy attention to their spatial configuration and to their capacity for becoming engines of economic growth before they become too locked in.

Fostering compact and efficient urban growth in these small cities and infrastructure that serves a growing population could help promote economic growth and productivity, avoiding the famed Africa’s urban development trap (Lall, Henderson, and Venables 2017), as well as promote equitable economic development with an eye to gender equality and climate resilience. Because shape operates through spatial reorganization rather than employment switching, policy interventions on urban form (compact and efficient urban growth, transportation and public utility infrastructure) can have outsized returns in small cities where the spatial equilibrium is not locked in yet, unlike the large cities where workers may have already adapted to the disamenities of urban living.

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A Appendix

A.1 Description of primary data survey modules

The primary survey collected detailed information across multiple topics not typically available in standard household surveys.

Basic socio-demographic information was collected (age, gender, marital status), along with ages of youngest and oldest children (when applicable) and child care modalities for children under 7. Respondents also provided a cursory residential history at the city level, indicating their state and LGA of birth, along with all the neighborhoods they had lived in in their current city in the past 10 years as well as all the cities they had lived in in the past 10 years. The survey also asked about reasons for choosing their current home location, with options including affordability, proximity to work, availability of better jobs, proximity to family, availability of housing, and safety. Respondents were also asked to report on the availability of basic services in their neighborhood: electricity, internet, garbage collection, water, public transportation.

The labor market module was created to hew as closely as possible to the LSMS employment module. The main labor market variables are an indicator for being active on the labor market, employment status (employed, self-employed, unemployed), hours worked in the past week on the main job, number of months worked in the past year, and an indicator for formal employment. Additional questions also collect typical weekly income or revenue for wage workers or self-employed workers respectively, work location (along with GPS information), work quality indicators such as benefits, work schedule as well as prospective questions to collect information on job dimensions that workers thought essential or problematic based on a list of 9 items (compensation, distance from home, hours, commuting time/cost, purpose of work, colleagues, management) or self-reported desire to move jobs.

The survey also contains novel data on commuting in Nigerian cities. Respondents reported their usual commute time (in minutes), weekly commute costs (in Naira), and primary mode of transportation (walking, bicycle, private motorcycle, private car, motorcycle taxi, keke napep, minibus, bus, taxi, or ride-hailing service). Additionally, respondents indicated whether they own any of the following transportation assets: a car, motorcycle, or bicycle. Later questions also emphasized accessibility to common destinations (worship locations, healthcare facilities, schools) by foot and by public transportation at the level of their neighborhood.

The safety module was a combination of questions adapted from Afrobarometer survey questionnaires and IFPRI's WEAI modules. It asked respondents to rate how safe they felt during their commute on a scale of 1 to 7, along with victimization self-reports of mugging, sexual harassment, or sexual assault. To capture perceptions of safety, respondents estimated the prevalence of these crimes affecting people of their own gender in their city. Finally,

respondents indicated whether they had experienced a non-fatal traffic accident in the past 3 months or knew someone who had been in a fatal traffic accident in the past 12 months.

A.2 Description and analysis of other outcomes

A.2.1 Gender, migration status and centrality

Beyond gender, I also examine the role of migration status, neighborhood characteristics and distance to city center in explaining response to city shape. I detail below the construction of these variables. The primary data included the collection of state and LGA of birth for all respondents, as well as residential history of past 10 years (which is not used here). At the time of the survey, 39% of respondents lived in their LGA of birth, and 68% in their state of birth. Past urban economics research in Sub-Saharan Africa does not distinguish for the most part between migrants and non-migrants, despite high prevalence of rural migrants and relatively high levels of migration compared to high-income countries' cities. This matters because migrants might be more likely and able to select where they live based on city shape, especially if city shape is strongly correlated with labor market outcomes, and thus should be more responsive to city shape.

Using information and preferences collected, I find that migrants and locals seem to make different residential decisions, with migrants being 17 percentage points (+44%) more likely to report choosing their home location because of affordability compared to 39% of locals, or 6 percentage points (+22%) more likely to choose based on proximity to jobs compared to 22% of locals, and 18 percentage points (-35%) less likely to choose their residential location based on family, as opposed to 52% of locals. Moreover, migrants seem to live in “better” neighborhoods, as denoted by more reliable access to water (+8 percentage points) and electricity (+6 percentage points), despite already high rates of access to water and electricity (62% and 93%, respectively) for locals. Despite all these differences, as I show below in Table A14, there is little distinction in how migrants and locals respond to city shape in this setting. A caveat to this analysis is that it does not take into account past migrations or migration history, which might influence where natives choose to locate if, for example, they have moved away from their LGA between birth and the date of the interview. However, clear differences in both preferences or self-reported neighborhood characteristics can help mitigate this concern.

As further exploration, I try to exploit within-city variation by comparing central versus peripheral LGAs, within the larger LSMS sample. I use Google's Open Buildings satellite data to compute quartiles and averages of building height at the LGA level to identify central versus peripheral LGAs. I denote as central the LGAs that have their average building height above the 75th percentile in height for their entire agglomeration. Cities with only one LGAs are counted as 0.

A.2.2 Neighborhood outcomes

I also try to investigate whether there is heterogeneity based on neighborhood-level characteristics that could help explain the labor results. I use the data collected to compute a standardized index of neighborhood quality, based on reported presence of utility infrastructure and of amenities at the neighborhood level. I leverage the information on safety and create a safety index using victimization self-reports for theft, mugging, sexual harassment, sexual assault and perceptions of violence occurrence for individuals of own gender. I use both the continuous indices or indicators for above/below median for both safety and neighborhood infrastructure indices and find no results. Infrastructure index is defined as the sum of all neighborhood-level indicators that take the value of 1 if the respondent indicates having access to the following services in their neighborhood: garbage collection, electricity, reliable electricity, piped water, reliable piped water. The services index is defined by the sum of all neighborhood-level indicators that take the value of 1 if the respondent indicates having access to the following services in their neighborhood: garbage collection, electricity, reliable electricity, piped water, reliable piped water, internet, public transportation, healthcare, schools.

A.3 Additional tables

A.3.1 Summary tables

Table A1: Sample comparison between LSMS and primary data

	primary data	LSMS	Combined
Female	0.48	0.51	0.50
Age	35.76	37.50	36.94
Married	0.60	0.53	0.55
Can read and write in English	0.90	0.72	0.79
Ever attended school	0.97	0.85	0.89
Completed years of education	12.95	9.56	10.66
Active on labor market this week	0.85	0.68	0.74
Searching for a job this week	0.06	0.04	0.04
Self-employed	0.43	0.31	0.35
Number of hours worked in past week	36.04	25.40	28.85
Formal job	0.20	0.09	0.13
Monthly wage (in Naira)	110634.52	86402.06	91133.48
Works in agricultural sector	0.05	0.20	0.14
Works in manufacturing sector	0.02	0.13	0.08
Works in low-productivity services	0.42	0.53	0.48
Works in services	0.26	0.14	0.20
Lives in city with <500k	0.42	0.42	0.42
Lives in city with 500k-1m people	0.16	0.21	0.19
Lives in city with 1m-5m people	0.25	0.19	0.21

Note: Sample means for the primary data collected for this study and the Nigeria Living Standards Survey (LSMS) 2018/2019. The primary data sample is restricted to labor force participants in 13 Nigerian cities, with respondents sampled 60% at home and 40% at transportation hubs. The LSMS sample is restricted to urban respondents in cities also covered by the primary data. “Combined” pools observations from both samples. “Formal” is defined as having access to employee-provided health insurance scheme, a pension or a written employment contract. Monthly wage is reported in Nigerian Naira.

Table A2: Summary statistics by sampling location, primary data sample

	Settlement	Transit hub	All
Female	0.49	0.47	0.48
Age	35.11	36.63	35.76
Completed years of education	13.01	12.88	12.95
Lives in state of birth	0.72	0.64	0.69
Lives in an informal settlement	0.51	0.45	0.48
Feels safe (1 to 7)	2.36	2.42	2.39
Searching for a job this week	0.06	0.05	0.06
Self-employed	0.44	0.42	0.43
Wage employed	0.24	0.27	0.25
Number of hours worked in past week	33.39	39.57	36.04
Average weekly income (in Naira)	25186.94	23869.79	24585.45
Average weekly revenue (in Naira)	29172.07	30395.82	29672.39
Work is same location as home	0.34	0.23	0.29
Respondent owns a transportation asset	0.25	0.20	0.22
Respondent owns a car	0.10	0.08	0.09
Respondent owns a motorcycle	0.11	0.10	0.10
Respondent owns a bicycle	0.02	0.01	0.02
Commute distance (in kms)	8.14	8.16	8.15
Commute time in mins	37.31	33.70	35.77
Commute cost (in Naira)	800.27	916.11	854.43
Weekly fuel cost (in Naira)	12097.12	13595.54	12650.99
<i>Primary commuting mode</i>			
Buses	14.07	23.95	18.69
Keke Napep	16.34	13.95	15.22
Minibuses	9.63	10.00	9.80
Motorcycle taxi	15.58	13.09	14.42
Other	3.79	1.11	2.54
Private bicycle	1.30	0.86	1.10
Private motorcycle	11.15	7.90	9.63
Private vehicle/car	7.68	6.30	7.04
Ride hailing	0.22	0.99	0.58
Taxi (car)	6.49	8.40	7.38
Walking	13.74	13.46	13.61
<i>Obs.</i>	1404	1058	2462

Note: Sample restricted to labor force participants, i.e. respondents who either held a job or were actively seeking one at the time of the survey. Respondents were sampled 60% at home (one per household) and 40% at transportation hubs, alternating between male and female respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. The safety scale ranges from 1 (very safe) to 7 (very unsafe). Weekly income and revenue are reported in Nigerian Naira and weekly revenue is only reported for those who report being self-employed. Weekly fuel cost is reported in Nigerian Naira. Commute distance, time, cost and fuel cost are winsorized at the 1% level.

Table A3: Robustness: Pre-determined geographic constraints

	Proximity	Cohesion	Dispersion	Spin	Summary
Elevation, m	-0.0003 (0.0009)	-0.0005 (0.0013)	-0.0008 (0.0014)	-0.0294 (0.0383)	-0.0001 (0.0002)
Distance from coast	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0024 (0.0031)	-0.0000 (0.0000)
Distance from water	-0.0000 (0.0001)	-0.0000 (0.0001)	-0.0001 (0.0001)	-0.0015 (0.0023)	-0.0000 (0.0000)
Distance from mineral deposit	-0.0004 (0.0021)	-0.0003 (0.0031)	0.0013 (0.0033)	0.0552 (0.0921)	0.0001 (0.0004)
Ruggedness, deg	-0.3834 (0.4956)	-0.5682 (0.7201)	-0.8706 (0.7443)	-19.7935 (20.4740)	-0.0949 (0.1013)
Bedrock depth, m	-0.0729 (0.0760)	-0.1043 (0.1060)	-0.1135 (0.1005)	-2.0787 (2.2322)	-0.0137 (0.0136)
Crop suitability, t/ha	0.2502 (0.8333)	0.2929 (1.1741)	0.0566 (1.1719)	7.2139 (24.6697)	0.0344 (0.1518)
Distance from deep water port	-0.0003 (0.0006)	-0.0005 (0.0009)	-0.0005 (0.0009)	-0.0174 (0.0233)	-0.0001 (0.0001)
Nearby ocean depth, m	0.0000 (0.0001)	0.0001 (0.0001)	0.0001 (0.0001)	0.0024 (0.0024)	0.0000 (0.0000)
<i>Obs.</i>	56	56	56	56	56

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Robust standard errors in parentheses. Sample includes cities in the LSMS analysis sample. Each cell reports the coefficient from a separate OLS regression of the shape metric (column) on the geographic characteristic (row), controlling for log city area. The summary index is the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Except when specified in the table, distances are in kilometers.

A.3.2 OLS tables

Table A5: OLS: City-level outcomes, primary data sample

<i>Panel A: Labor market outcomes</i>						
	(1) Self-emp.	(2) Wage emp.	(3) Unemp.	(4) Hours	(5) Formal	(6) Same loc.
<i>Summary</i>	-0.009 (0.053)	-0.039 (0.061)	0.013 (0.015)	7.077 (4.407)	-0.023 (0.049)	0.081** (0.036)

	Self-emp.	Wage emp.	Unemp.	Hours	Formal	Same loc.	
R ²	0.002	0.025	0.015	0.093	0.015	0.400	
N	13	13	13	13	13	13	
Panel B: Other work outcomes							
	Log(wage)	Log(rev.)	Manuf.	Services	Low-prod.		
Summary	-0.092 (0.109)	-0.057 (0.184)	0.046 (0.028)	0.028 (0.026)	0.061 (0.061)		
R ²	0.021	0.003	0.146	0.039	0.048		
N	13	13	12	12	12		
Panel C: Firm size							
	1-person	Micro	Small	Medium	Large	Avg. size	
Summary	-0.032 (0.022)	-0.069* (0.034)	0.054 (0.032)	0.011 (0.009)	0.004*** (0.001)	1.474* (0.687)	
R ²	0.064	0.259	0.209	0.086	0.275	0.211	
N	13	13	13	13	13	13	
Panel D: Commuting							
	Time	Distance	Safety	Cost	Walk	Public	Informal
Summary	1.169 (8.333)	-1.921 (1.560)	0.074 (0.121)	271.875 (171.230)	0.015 (0.026)	0.054 (0.107)	-0.056 (0.054)
R ²	0.001	0.060	0.015	0.122	0.015	0.033	0.041
N	13	13	13	13	13	13	13

Note: *** p< 0.01, ** p< 0.05, * p< 0.10. Each cell reports the coefficient from a separate OLS regression of a city-level outcome on the *Summary* index, using the primary data sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values of the index indicate worse city shapes. All dependent variables are city-level averages. Panel A: “Active” means active on the labor market in the past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers and null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. “Same loc.” is defined as reporting working from the same address as one’s home location. Panel B: Wages are included only for non-self-employed respondents, and revenues for self-employed respondents exclusively; both are winsorized at the 1% level. Sectoral shares (manufacturing, services, low-productivity) are reconstructed from the share of LSMS respondents in each city and round working in a given sector: services (financial, education/health, government), low-productivity services (food, transportation, commerce, construction), and manufacturing and resource sectors. Panel C: Firm size categories are constructed from the Nigerian Business Census 2023 round and defined utilizing the World Bank thresholds for enterprise size: 1-person (sole operator), micro (5 or fewer), small (5-19), medium (20-99) and large (100+). “Avg. size” is the mean number of employees. Panel D: Commuting time is measured in minutes, distance in kilometers, and cost in local currency. “Walk,” “Public” and “Informal” denote the share of respondents using public or informal transit respectively as their primary commuting method. The safety variable is a Likert scale of respondents reporting how safe they felt during their commute, from a scale from 1 to 7, with 7 being the least safe.

Table A4: OLS: Commuting outcomes excluding Lagos, primary data sample

	(1) Time	(2) Distance	(3) Safety	(4) Cost	(5) Walk	(6) Public	(7) Informal
Proximity	7.939 (9.804)	1.632 (2.601)	0.141 (0.148)	69.869 (136.278)	-0.052* (0.027)	0.001 (0.062)	-0.023 (0.076)
Cohesion	6.323 (7.093)	1.526 (1.858)	0.098 (0.101)	65.185 (95.419)	-0.039* (0.019)	0.010 (0.043)	-0.020 (0.053)
Spin	0.111 (0.206)	0.057 (0.085)	0.005 (0.005)	3.197 (5.466)	-0.002* (0.001)	0.001 (0.002)	-0.002 (0.002)
Dispersion	0.538 (3.335)	0.952 (2.332)	0.114 (0.112)	18.117 (72.695)	-0.035 (0.023)	0.031 (0.039)	-0.056 (0.035)
<i>Summary</i>	32.650 (44.721)	10.038 (15.799)	0.860 (0.827)	400.182 (765.790)	-0.305* (0.162)	0.104 (0.333)	-0.271 (0.364)
Observations	1945	2021	2039	1462	1470	1470	1470

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate OLS regression of the outcome (column) on the shape (row) and includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population using the primary data sample. This specification excludes all Lagos respondents. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values in the shape metrics indicate worse city shapes. The safety variable is a Likert scale of respondents reporting how safe they felt during their commute, from a scale from 1 to 7, with 7 being the least safe. The commuting time, distance and cost variables are winsorized at the top 1% percent. Distance is expressed in kilometers, time is expressed in minutes and costs are expressed in Nigerian Naira. “Public” refers to public transportation, defined as commuting by bus or minibus, and “Informal” to informal transportation defined as commuting by motorcycle or tricycle taxis.

Table A6: Robustness: OLS, LSMS sample

	(1)	(2)	(3)	(4)	(5)
	Active	Self-emp.	Unemp.	Hours	Formal
<i>Panel A: No controls</i>					
Cohesion	-0.002 (0.002)	-0.005** (0.002)	-0.000 (0.001)	0.296* (0.151)	0.005*** (0.001)
Proximity	-0.003 (0.002)	-0.008** (0.003)	0.000 (0.001)	0.422* (0.222)	0.007*** (0.002)
Spin	-0.000 (0.000)	-0.000*** (0.000)	-0.000 (0.000)	0.014*** (0.005)	0.000*** (0.000)
Dispersion	-0.002 (0.002)	-0.005** (0.002)	-0.000 (0.001)	0.314** (0.144)	0.005*** (0.001)
<i>Summary</i>	-0.017 (0.013)	-0.042*** (0.016)	-0.000 (0.007)	2.356** (1.068)	0.040*** (0.010)
<i>Obs.</i>	7136	7136	7136	5983	3754
<i>Panel B: Controlling for manufacturing share</i>					
Cohesion	-0.003* (0.002)	-0.004** (0.002)	-0.000 (0.001)	0.223 (0.145)	0.002* (0.001)
Proximity	-0.004* (0.002)	-0.005** (0.002)	0.000 (0.001)	0.315 (0.212)	0.003* (0.002)
Spin	-0.000 (0.000)	-0.000*** (0.000)	-0.000 (0.000)	0.011** (0.005)	0.000* (0.000)
Dispersion	-0.003* (0.002)	-0.003** (0.002)	-0.000 (0.001)	0.240* (0.137)	0.002* (0.001)
<i>Summary</i>	-0.022* (0.012)	-0.029** (0.011)	-0.000 (0.007)	1.814* (1.012)	0.017* (0.009)
<i>Obs.</i>	7129	7129	7129	5976	3747
<i>Panel C: Excluding cities with colonial origins</i>					
Cohesion	-0.003 (0.002)	-0.006** (0.002)	-0.001 (0.001)	0.330*** (0.106)	0.002* (0.001)
Proximity	-0.004 (0.003)	-0.008** (0.003)	-0.001 (0.001)	0.475*** (0.155)	0.003* (0.002)
Spin	-0.000 (0.000)	-0.000** (0.000)	-0.000 (0.000)	0.014*** (0.004)	0.000* (0.000)
Dispersion	-0.003 (0.002)	-0.005** (0.002)	-0.001 (0.001)	0.334*** (0.102)	0.002* (0.001)

	(1)	(2)	(3)	(4)	(5)
	Active	Self-emp.	Unemp.	Hours	Formal
<i>Summary</i>	-0.022 (0.014)	-0.042** (0.017)	-0.006 (0.005)	2.494*** (0.761)	0.016* (0.009)
<i>Obs.</i>	6096	6096	6096	5088	3143

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Estimates from an OLS specification of outcomes (column) on city shape metrics (row) using the LSMS sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Log of linearly projected population is included. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Panel A excludes controls for age, gender and education. Panel B adds city-level manufacturing share, reconstructed from 2023 LSMS corresponding to the share of respondents reporting their sector of activity, along with the usual controls for age, gender and years of education and correlates for log of linearly projected population. Panel C drops cities matching Baruah, Henderson, and Peng (2021)'s dataset, which identifies which cities were created by colonial powers.

Table A7: Robustness: OLS, primary data sample

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage emp.	Unemp.	Hours	Formal	Same loc.
<i>Panel A: No controls</i>						
Proximity	0.031** (0.013)	-0.008 (0.015)	0.002 (0.005)	2.797** (1.357)	-0.014 (0.013)	-0.001 (0.008)
Cohesion	0.022** (0.009)	-0.005 (0.010)	0.001 (0.003)	1.950** (0.929)	-0.010 (0.009)	-0.001 (0.006)
Spin	0.001*** (0.000)	-0.000 (0.000)	0.000 (0.000)	0.082** (0.030)	-0.000* (0.000)	-0.000 (0.000)
Dispersion	0.021** (0.008)	-0.006 (0.009)	0.002 (0.003)	1.814** (0.875)	-0.010 (0.008)	-0.001 (0.005)
<i>Summary</i>	0.169** (0.065)	-0.034 (0.070)	0.011 (0.024)	15.165** (6.676)	-0.079 (0.058)	-0.015 (0.042)
<i>Obs.</i>	2462	2462	2462	2462	2462	2462
<i>Panel B: Controlling for manufacturing share</i>						
Proximity	0.017 (0.011)	-0.000 (0.012)	-0.000 (0.006)	2.149 (1.481)	0.001 (0.012)	0.001 (0.007)
Cohesion	0.012 (0.008)	0.000 (0.008)	-0.000 (0.004)	1.500 (1.021)	0.001 (0.008)	0.000 (0.005)
Spin	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.065* (0.037)	0.000 (0.000)	-0.000 (0.000)
Dispersion	0.011 (0.008)	-0.001 (0.008)	0.000 (0.004)	1.324 (0.971)	0.000 (0.008)	0.000 (0.005)

Table A7: Robustness: OLS, primary data sample (continued)

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage emp.	Unemp.	Hours	Formal	Same loc.
<i>Summary</i>	0.090 (0.062)	0.010 (0.062)	-0.002 (0.032)	11.655 (7.609)	0.005 (0.061)	-0.004 (0.038)
<i>Obs.</i>	2362	2362	2362	2362	2362	2362
<i>Panel C: Excluding cities with colonial origins</i>						
Proximity	0.034** (0.014)	-0.013 (0.012)	0.006 (0.004)	2.295 (1.478)	-0.025*** (0.007)	-0.007 (0.009)
Cohesion	0.023** (0.009)	-0.009 (0.008)	0.004 (0.003)	1.588 (1.007)	-0.017*** (0.005)	-0.005 (0.006)
Spin	0.001** (0.000)	-0.000 (0.000)	0.000* (0.000)	0.069** (0.032)	-0.001*** (0.000)	-0.000 (0.000)
Dispersion	0.022** (0.009)	-0.008 (0.008)	0.004* (0.002)	1.530 (0.925)	-0.015*** (0.004)	-0.005 (0.006)
<i>Summary</i>	0.169** (0.068)	-0.056 (0.059)	0.032* (0.018)	12.515* (7.148)	-0.119*** (0.033)	-0.040 (0.043)
<i>Obs.</i>	2057	2057	2057	2057	2057	2057

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Estimates from a OLS specification of outcomes (column) on city shape metrics (row) using the LSMS sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Log of linearly projected population is included. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Panel A excludes controls for age, gender and education. Panel B adds city-level manufacturing share, reconstructed from 2023 LSMS corresponding to the share of respondents reporting their sector of activity, along with the usual controls for age, gender and years of education and correlates for log of linearly projected population. Panel C drops cities matching Baruah, Henderson, and Peng (2021)'s dataset, which identifies which cities were created by colonial powers.

A.3.3 IV tables

Table A8: IV: First stage, both samples

	<i>LSMS sample</i>				
	(1) Proximity	(2) Cohesion	(3) Spin	(4) Dispersion	(5) Summary
Pred. Cohesion	1.068*** (0.091)				
Pred. Proximity		1.017*** (0.083)			
Pred. Spin			0.887*** (0.045)		
Pred. Dispersion				1.088*** (0.118)	
Pred. Summary					1.195*** (0.094)
F-statistic	90.48	110.34	468.06	37.54	104.64
	<i>Primary data sample</i>				
	(1)	(2)	(3)	(4)	(5)
Pred. Proximity	0.970*** (0.154)				
Pred. Cohesion		1.023*** (0.161)			
Pred. Spin			1.034*** (0.090)		
Pred. Dispersion				1.091*** (0.182)	
Pred. Summary					1.231*** (0.166)
F-statistic	71.79	67.01	198.83	54.89	84.59

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level in the top panel and at the LGA level in the bottom panel. Each coefficient is the coefficient for the first-stage of a 2SLS specification between shape and labor outcomes instrumented with predicted shape metrics and includes controls for age, gender and number of years of education, as well as log of linearly projected population. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Predicted metrics are the computed values of the metrics for the city shapes that are within the 2025 expanded bounding circle, based on 1975 actual city shape as the instrument.

A.3.4 IV Robustness specifications

Table A9: IV: Other labor outcomes, LSMS sample

	(1) Log(wage)	(2) Log(revenue)	(3) Manufacturing	(4) Services	(5) Low-prod. services	(6) Log(wage)
Cohesion	-0.018 (0.039)	-0.034 (0.055)	0.005 (0.040)	-0.018** (0.009)	0.007 (0.008)	-0.026 (0.022)
Proximity	-0.026 (0.058)	-0.049 (0.080)	0.008 (0.061)	-0.027** (0.013)	0.010 (0.012)	-0.040 (0.033)
Spin	0.000 (0.001)	-0.000 (0.001)	0.000 (0.001)	-0.000* (0.000)	0.000 (0.000)	-0.000 (0.000)
Dispersion	-0.050 (0.076)	-0.077 (0.110)	-0.014 (0.067)	-0.032 (0.020)	0.010 (0.013)	-0.043 (0.041)
<i>Summary</i>	-0.097 (0.236)	-0.199 (0.327)	0.045 (0.249)	-0.110** (0.054)	0.039 (0.049)	-0.163 (0.135)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape using the LSMS sample. The specification also includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population and a year dummy. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Wages are only included for non self-employed respondents, and revenues for self-employed respondents exclusively. Both wages and revenue are winsorized at the 1% level. Sectors are defined as follows: services (financial, education/health, govt), low-productivity services (food, transportation, commerce, construction) and manufacturing and resource sectors. Each variable is a dummy for working in this sector of activity. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Table A10: IV: Firm size (collapsed), LSMS sample

	(1)	(2)	(3)	(4)	(5)	(6)
	1-person	Micro	Small	Medium	Large	Firm size
Proximity	0.033** (0.014)	0.010 (0.009)	-0.005 (0.008)	-0.005 (0.004)	0.001** (0.000)	-0.162 (0.162)
Cohesion	0.023** (0.010)	0.007 (0.007)	-0.004 (0.006)	-0.004 (0.003)	0.001** (0.000)	-0.115 (0.114)
Spin	0.001 (0.001)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000* (0.000)	-0.005 (0.005)
Dispersion	0.024** (0.010)	0.008 (0.007)	-0.005 (0.006)	-0.004 (0.003)	0.000** (0.000)	-0.147 (0.120)
<i>Summary</i>	0.177** (0.081)	0.051 (0.050)	-0.027 (0.043)	-0.028 (0.023)	0.005** (0.002)	-0.941 (0.866)
Control mean	0.19	0.57	0.36	0.07	0.002	7.02
<i>Obs. (cities)</i>	51	51	51	51	51	51

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each cell reports the coefficient from a separate 2SLS regression of a city-level outcome (column) on the shape metric (row), using the LSMS sample. Each specification includes controls for city-level averages of age, gender and education as well as log of linearized projected population. All dependent variables are city-level averages. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Firm size categories are constructed from the Nigerian Business Census 2023 round and defined utilizing the World Bank thresholds for enterprise size: 1-person (sole operator), micro (5 or fewer), small (5-19), medium (20-99) and large (100+). “Firm size” is the mean number of employees. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Table A11: IV: Heterogeneity by gender, primary data sample

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage empl	Unemp.	Nb. Hours	Formal	Same loc.
<i>Summary</i>	0.686 (0.442)	-0.383 (0.357)	0.033 (0.062)	6.864 (26.196)	-0.415* (0.239)	-0.286* (0.168)
Fem. x <i>Summary</i>	-0.009 (0.033)	0.031 (0.031)	-0.006 (0.018)	3.929 (2.435)	-0.004 (0.021)	0.043 (0.030)
Female	0.080** (0.033)	0.024 (0.034)	-0.016 (0.013)	1.783 (2.119)	-0.010 (0.028)	0.050** (0.023)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape using the primary data sample. Controls for age and number of years of education, and log of linearly projected population are included. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Higher values in the *Summary* index indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. “Same location” is defined as reporting working from the same address as one’s home location.

Table A12: Heterogeneity by gender and city size: remaining outcomes, LSMS sample

	Unemployed		Hours worked		Formal	
	Small (<300k)	Large (≥300k)	Small (<300k)	Large (≥300k)	Small (<300k)	Large (≥300k)
<i>Summary</i>	-0.0679 (0.1290)	-0.0032 (0.0033)	-15.8621 (12.7809)	1.0319 (0.6645)	0.1002 (0.1261)	0.0390*** (0.0055)
Female	-0.0115 (0.0671)	-0.0063 (0.0056)	4.4160 (8.7608)	-3.1307*** (1.0630)	-0.0720 (0.0826)	-0.0041 (0.0136)
Female × <i>Summary</i>	-0.0075 (0.0862)	0.0084*** (0.0024)	11.1557 (12.8585)	0.8524 (0.5445)	-0.0915 (0.1098)	-0.0501*** (0.0079)
Male control mean	0.0504	0.0431	29.3338	31.2539	0.0727	0.1293
[<i>small</i> = <i>large</i>] p-value	0.852		0.419		0.704	
<i>Obs.</i>	1932	5204	1585	4398	935	2819

Note: *** p < 0.01, ** p < 0.05, * p < 0.10. Clustered standard errors at the city level. Estimates from a 2SLS specification where 2025 city shape metrics are instrumented using 1975 actual city shape with the LSMS sample. Each column corresponds to a regression of the self-employed indicator on the cohesion or the summary metric interacted with the female dummy, separately for cities with population below and above 300,000 people. Controls for age and number of years of education are included, as well as year dummy. City population is taken from the UN World Urbanization Prospects data and from citypopulation.de. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract and is only collected in 2023 LSMS. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Table A13: IV: Heterogeneity by gender (other outcomes), LSMS sample

	(1) Income (log)	(2) Wages (log)	(3) Revenue (log)
<i>Summary</i>	-0.097 (0.238)	-0.179 (0.338)	0.013 (0.254)
Female x <i>Summary</i>	-0.000 (0.039)	-0.037 (0.041)	0.053 (0.065)
Female	-0.595*** (0.054)	-0.467*** (0.072)	-0.747*** (0.074)
<i>Obs.</i>	3154	1544	1610

Note: *** p < 0.01, ** p < 0.05, * p < 0.10. Clustered standard errors at the city level. Each cell reports the coefficient from a separate 2SLS specification outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape using the LSMS sample. The specification also includes controls for age, gender and number of years of education for each respondent, as well as log of linearly projected population and a year dummy. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Wages are only included for non self-employed respondents, and revenues for self-employed respondents exclusively. Both wages and revenue are winsorized at the 1% level. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Table A14: IV: Heterogeneity by migration status, primary data sample

	(1) Self-emp.	(2) Wage empl	(3) Unemp.	(4) Nb. Hours	(5) Formal	(6) Same loc.
<i>Summary</i>	-0.033 (0.045)	-0.000 (0.033)	0.026* (0.014)	7.714** (3.223)	-0.053** (0.027)	0.058 (0.036)
LGA native x <i>Summary</i>	-0.096 (0.060)	0.010 (0.034)	-0.001 (0.010)	-7.682 (5.065)	0.080** (0.034)	0.040 (0.046)
LGA native	-0.013 (0.054)	-0.079** (0.035)	-0.005 (0.009)	-6.511 (4.806)	0.013 (0.037)	0.038 (0.032)

Note: *** p< 0.01, ** p< 0.05, * p< 0.10. Clustered standard errors at the LGA level. Controls for age and number of years of education, and log of linearly projected population are included. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). LGA natives are respondents who currently live in the same LGA where they were born. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. “Same location” is defined as reporting working from the same address as one’s home location.

Table A17: Robustness: IV, primary data sample

	(1) Self-emp.	(2) Wage empl	(3) Unemp.	(4) Hours	(5) Formal	(6) Same loc.
<i>Panel A: No controls</i>						
Proximity	0.205 (0.170)	-0.124 (0.143)	0.004 (0.017)	1.496 (7.586)	-0.134 (0.105)	-0.063 (0.047)
Cohesion	0.133 (0.105)	-0.079 (0.089)	0.004 (0.011)	1.057 (4.966)	-0.087 (0.066)	-0.042 (0.030)
Spin	0.002*** (0.001)	-0.001 (0.001)	-0.000 (0.000)	0.062 (0.060)	-0.001*** (0.000)	-0.001* (0.000)
Dispersion	0.252 (0.457)	-0.221 (0.432)	0.014 (0.021)	-2.569 (11.597)	-0.158 (0.280)	-0.072 (0.110)
<i>Summary</i>	0.768 (0.501)	-0.452 (0.441)	0.015 (0.064)	8.348 (26.883)	-0.477 (0.299)	-0.238 (0.154)
<i>Panel B: Controlling for manufacturing share</i>						
Proximity	0.500 (2.608)	-0.615 (3.092)	-0.162 (0.740)	-99.046 (482.948)	-0.158 (0.814)	-0.333 (1.534)
Cohesion	0.273 (1.211)	-0.329 (1.404)	-0.082 (0.315)	-55.140 (226.217)	-0.087 (0.383)	-0.192 (0.745)
Spin	0.001 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.040 (0.121)	-0.000 (0.000)	-0.001 (0.001)
Dispersion	-0.034	0.113	0.010	14.897	-0.002	0.023

	(1)	(2)	(3)	(4)	(5)	(6)
	Self-emp.	Wage empl	Unemp.	Hours	Formal	Same loc.
<i>Summary</i>	(0.056)	(0.078)	(0.026)	(11.631)	(0.025)	(0.046)
	0.922	-1.128	-0.303	-174.393	-0.138	-0.641
	(2.305)	(2.575)	(0.631)	(378.530)	(0.558)	(1.305)
<i>Panel C: Excluding cities with colonial origins</i>						
Proximity	0.185	-0.096	0.009	3.155	-0.103	-0.073
	(0.155)	(0.120)	(0.011)	(7.284)	(0.081)	(0.050)
Cohesion	0.118	-0.058	0.006	2.202	-0.066	-0.047
	(0.092)	(0.072)	(0.007)	(4.630)	(0.049)	(0.031)
Spin	0.002***	-0.001	0.000	0.064	-0.001***	-0.001**
	(0.001)	(0.001)	(0.000)	(0.072)	(0.000)	(0.000)
Dispersion	0.174	-0.126	0.005	0.529	-0.087	-0.059
	(0.233)	(0.194)	(0.011)	(6.573)	(0.118)	(0.063)
<i>Summary</i>	0.709	-0.368	0.042	13.514	-0.385	-0.281*
	(0.472)	(0.393)	(0.043)	(27.022)	(0.247)	(0.164)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Estimates from a 2SLS specification of outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape, using the primary data sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Log of linearly projected population is included. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values of the shape metrics indicate worse city shapes. Panel A excludes controls for age, gender and education. Panel B adds city-level manufacturing share, reconstructed from 2023 LSMS corresponding to the share of respondents reporting their sector of activity, along with the usual controls for age, gender and years of education and correlates for log of linearly projected population. Panel C drops cities matching Baruah, Henderson, and Peng (2021)'s dataset, which identifies three cities in the sample, which were created by colonial powers.

Table A18: Robustness: IV, LSMS sample

	(1)	(2)	(3)	(4)	(5)
	Active	Self-emp.	Unemp.	Hours	Formal
<i>Panel A: No controls</i>					
Proximity	-0.014	-0.039*	-0.008	1.363	0.007
	(0.015)	(0.020)	(0.006)	(0.877)	(0.007)
Cohesion	-0.009	-0.026*	-0.005	0.923	0.005
	(0.010)	(0.014)	(0.004)	(0.589)	(0.005)
Spin	-0.000	-0.000**	-0.000	0.020*	0.000
	(0.000)	(0.000)	(0.000)	(0.011)	(0.000)
Dispersion	-0.013	-0.037	-0.010	1.237	0.004
	(0.016)	(0.026)	(0.008)	(0.916)	(0.006)

	(1)	(2)	(3)	(4)	(5)
	Active	Self-emp.	Unemp.	Hours	Formal
<i>Summary</i>	-0.052 (0.061)	-0.160* (0.085)	-0.034 (0.024)	5.906 (3.613)	0.034 (0.030)
<i>Panel B: Controlling for manufacturing share</i>					
Proximity	-0.013 (0.014)	-0.020 (0.016)	-0.008 (0.006)	1.266 (0.774)	0.001 (0.006)
Cohesion	-0.009 (0.009)	-0.013 (0.010)	-0.005 (0.004)	0.848* (0.515)	0.001 (0.004)
Spin	-0.000 (0.000)	-0.000* (0.000)	-0.000 (0.000)	0.019** (0.009)	0.000 (0.000)
Dispersion	-0.013 (0.016)	-0.016 (0.018)	-0.009 (0.008)	1.166 (0.806)	-0.000 (0.005)
<i>Summary</i>	-0.047 (0.057)	-0.082 (0.063)	-0.032 (0.023)	5.442* (3.144)	0.006 (0.026)
<i>Panel C: Excluding cities with colonial origins</i>					
Proximity	-0.019 (0.014)	-0.041** (0.020)	-0.007 (0.005)	0.910 (0.723)	0.003 (0.006)
Cohesion	-0.013 (0.009)	-0.027** (0.014)	-0.005 (0.004)	0.622 (0.486)	0.002 (0.004)
Spin	-0.000 (0.000)	-0.001** (0.000)	-0.000 (0.000)	0.017* (0.010)	0.000 (0.000)
Dispersion	-0.016 (0.013)	-0.036* (0.022)	-0.007 (0.005)	0.667 (0.632)	0.002 (0.004)
<i>Summary</i>	-0.079 (0.059)	-0.174** (0.086)	-0.030 (0.022)	4.127 (3.073)	0.016 (0.026)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level. Estimates from a 2SLS specification of outcomes (column) on 2025 city shape metrics (row), instrumented using shape metrics of the predicted expansion of the 1975 actual city shape, using the LSMS sample. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Log of linearly projected population and a year dummy are included. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values of the shape metrics indicate worse city shapes. Panel A excludes controls for age, gender and education. Panel B adds city-level manufacturing share, reconstructed from LSMS corresponding to the share of respondents reporting their sector of activity, along with the usual controls for age, gender and years of education and correlates for log of linearly projected population. Panel C drops cities matching Baruah, Henderson, and Peng (2021)'s dataset, which identifies three cities in the sample, which were created by colonial powers. Missing values for years of education are imputed and an indicator variable for missing observations is included in the regression.

Table A15: IV: Neighborhood quality outcomes, primary data sample

	Self-emp. (1)	Wage empl (2)	Unemp. (3)	Hours (4)	Formal (5)	Same loc. (6)
<i>Panel A: Neighborhood infrastructure</i>						
<i>Summary</i>	-0.056 (0.046)	0.002 (0.030)	0.025** (0.010)	5.183 (3.228)	-0.030 (0.021)	0.073** (0.030)
Neighborhood infrastructure $\times$ <i>Summary</i>	-0.011 (0.008)	-0.001 (0.006)	0.004 (0.003)	-0.125 (0.683)	-0.005 (0.003)	-0.005 (0.004)
Neighborhood infrastructure	-0.012 (0.008)	0.005 (0.007)	-0.001 (0.003)	0.147 (0.700)	0.020*** (0.004)	0.001 (0.004)
<i>Panel B: Safety</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Summary</i>	-0.054 (0.043)	-0.002 (0.028)	0.026** (0.012)	5.227 (3.232)	-0.026 (0.023)	0.072*** (0.027)
Safety index $\times$ <i>Summary</i>	-0.044*** (0.010)	0.013 (0.008)	0.003 (0.004)	-0.601 (1.108)	0.007 (0.008)	-0.011 (0.010)
Safety index	-0.009 (0.008)	0.011** (0.005)	-0.004 (0.003)	0.262 (0.727)	-0.005 (0.006)	0.003 (0.007)
<i>Panel C: Neighborhood services</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Summary</i>	-0.053 (0.045)	0.000 (0.029)	0.025** (0.011)	5.004 (3.199)	-0.031 (0.023)	0.074*** (0.029)
Neighborhood services $\times$ <i>Summary</i>	-0.011** (0.005)	0.001 (0.004)	0.001 (0.002)	0.135 (0.431)	-0.001 (0.002)	-0.003 (0.003)
Neighborhood services	-0.006 (0.005)	0.005 (0.004)	-0.000 (0.001)	0.142 (0.429)	0.012*** (0.002)	-0.000 (0.003)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the LGA level. Each cell reports the coefficient from a separate 2SLS specification of labor outcomes (column) on 2025 city shape metrics (row), instrumented using 1975 actual city shape. Controls for age, gender and years of education are also included. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). The neighborhood infrastructure index is the standardized sum of all neighborhood-level indicators that take the value of 1 if the respondent indicates having access to the following services in their neighborhood: garbage collection, electricity, reliable electricity, piped water, reliable piped water. The neighborhood services index is the standardized sum of all neighborhood-level indicators that take the value of 1 if the respondent indicates having access to the following services in their neighborhood: garbage collection, electricity, reliable electricity, piped water, reliable piped water, internet, public transportation, healthcare, schools. The safety index is the standardized sum of all three indicators for ever having been a victim of mugging, sexual harassment or sexual assault; higher values indicate worse safety outcomes, recoded so that positive values indicate better safety outcomes. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values of the metrics and *Summary* indicate worse city shapes. “Active” means active on labor market this past week, including both employed and job-seeking respondents. Hours worked are set to zero for unemployed workers, null for inactive respondents. “Formal” is defined as having access to a health insurance scheme, a pension or a written employment contract. “Same location” is defined as reporting working from the same address as one’s home location.

Table A16: Robustness: First stage excluding controls, both samples

	<i>LSMS sample</i>				
	(1) Proximity	(2) Cohesion	(3) Spin	(4) Dispersion	(5) Summary
Pred. Cohesion	1.063*** (0.093)				
Pred. Proximity		1.012*** (0.086)			
Pred. Spin			0.883*** (0.045)		
Pred. Dispersion				1.082*** (0.120)	
Pred. Summary					1.188*** (0.096)
F-statistic	99.05	107.21	252.85	55.86	108.10
	<i>Primary data sample</i>				
	(1)	(2)	(3)	(4)	(5)
Pred. Proximity	0.964*** (0.160)				
Pred. Cohesion		1.017*** (0.167)			
Pred. Spin			1.028*** (0.092)		
Pred. Dispersion				1.079*** (0.190)	
Pred. Summary					1.221*** (0.172)
F-statistic	87.90	87.43	261.71	67.03	102.13

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered standard errors at the city level in the top panel and at the LGA level in the bottom panel. Each coefficient is the coefficient for the first-stage of a 2SLS specification between shape and labor outcomes instrumented with predicted shape metrics and includes controls for age, gender and number of years of education, as well as log of linearly projected population. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). The predicted *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion) for predicted city shapes in 2025, i.e. the developable city shapes that are contained within the 2025 expanded bounding circle, based on 1975 actual city shape as the instrument. Other predicted outcomes are constructed in an analogous manner. Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Predicted metrics are the computed values of the metrics for the city shapes that are within the 2025 expanded bounding circle, based on 1975 actual city shape as the instrument.

Table A19: IV: Firm size excluding controls, LSMS sample

	(1)	(2)	(3)	(4)	(5)	(6)
	1-person	Micro	Small	Medium	Large	Firm size
Proximity	-0.013 (0.010)	-0.014* (0.008)	0.015* (0.008)	-0.002 (0.003)	0.001*** (0.000)	0.340** (0.152)
Cohesion	-0.009 (0.007)	-0.010* (0.006)	0.010* (0.006)	-0.001 (0.002)	0.001*** (0.000)	0.241** (0.109)
Spin	-0.000 (0.000)	-0.001 (0.000)	0.001 (0.000)	-0.000 (0.000)	0.000** (0.000)	0.012* (0.007)
Dispersion	-0.008 (0.007)	-0.009 (0.006)	0.010* (0.006)	-0.001 (0.002)	0.001*** (0.000)	0.217** (0.100)
<i>Summary</i>	-0.070 (0.055)	-0.079* (0.048)	0.082* (0.048)	-0.010 (0.017)	0.006*** (0.002)	1.847** (0.872)
Control mean	0.19	0.57	0.36	0.07	0.002	7.02
<i>Obs.</i> (cities)	51	51	51	51	51	51

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each cell reports the coefficient from a separate 2SLS regression of a city-level outcome (column) on the shape metric (row), using the LSMS sample. Each specification includes only log of linearized projected population. All dependent variables are city-level averages. *Summary* is computed as the standardized first component from a principal component analysis of all the shape metrics (proximity, cohesion, spin, dispersion). Proximity, dispersion and cohesion are expressed in kilometers; spin is expressed in square kilometers. Higher values in the shape metrics indicate worse city shapes. Firm size categories are constructed from the Nigerian Business Census 2023 round and defined utilizing the World Bank thresholds for enterprise size: 1-person (sole operator), micro (5 or fewer), small (5-19), medium (20-99) and large (100+). “Firm size” is the mean number of employees.

B Additional figures

Figure A1: Google Maps and city shapes: distance and duration for trips to city center

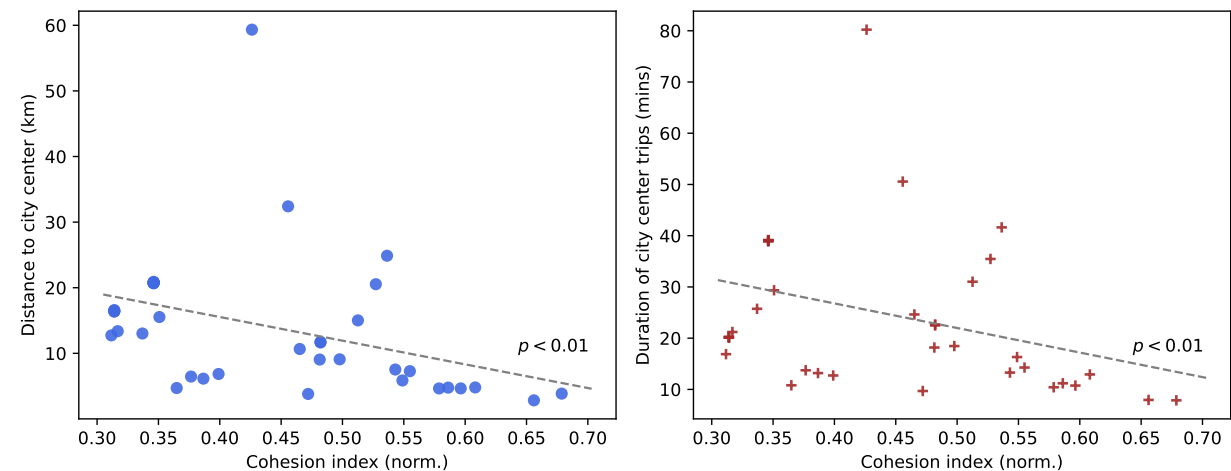
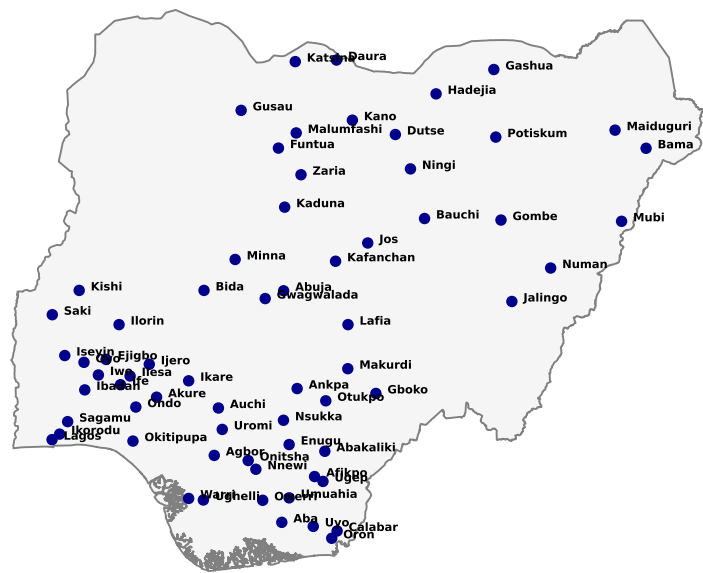


Figure A2: Sample cities



(a) LSMS



(b) Primary data

Figure A3: City shape in 1975 for the primary data sample

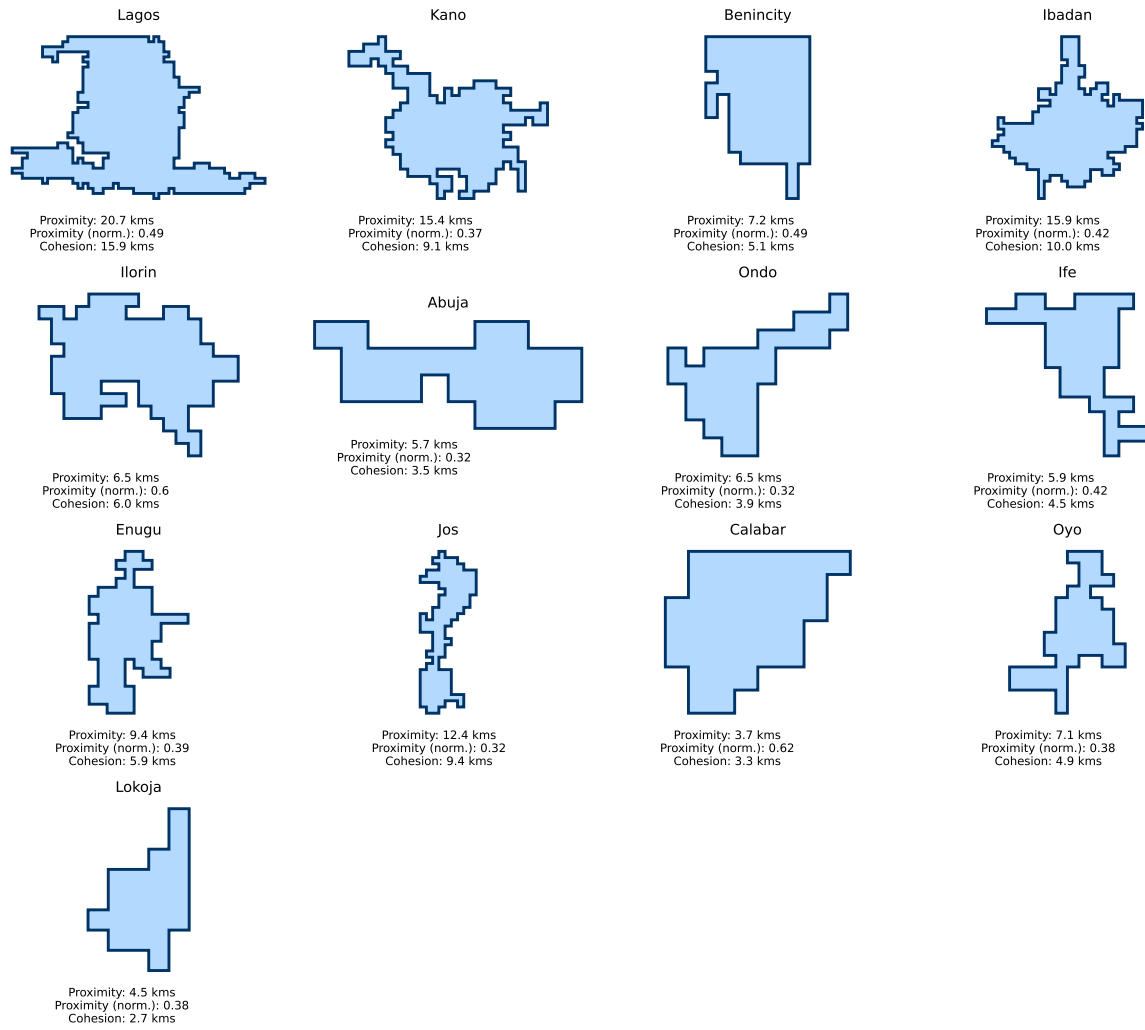


Figure A4: Summary index: percentile distribution

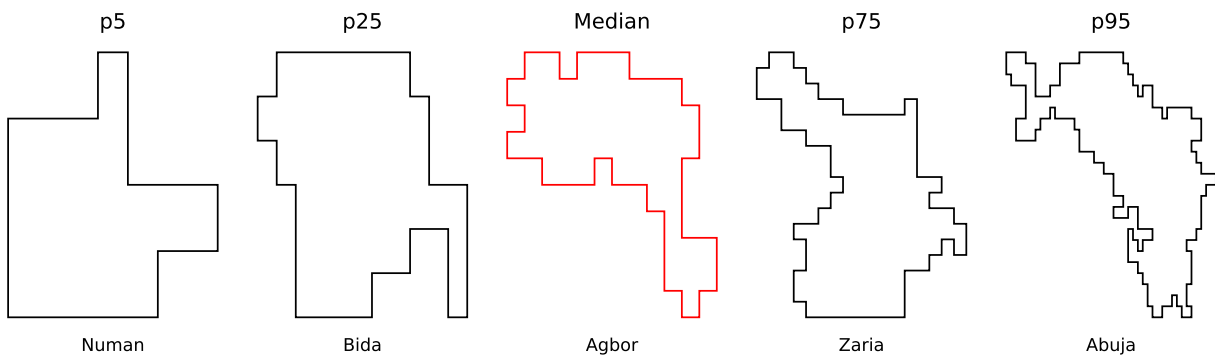


Figure A5: Firm size and city shape: correlation

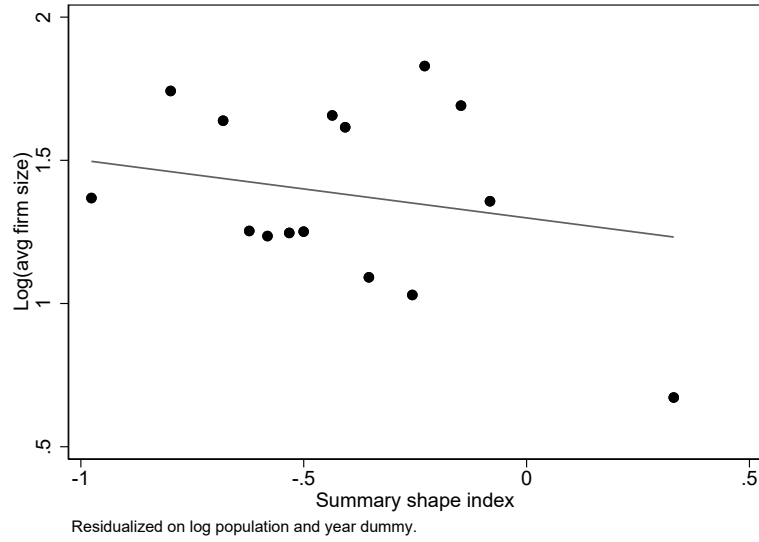


Figure A6: Distribution of job locations by distance to the CBD

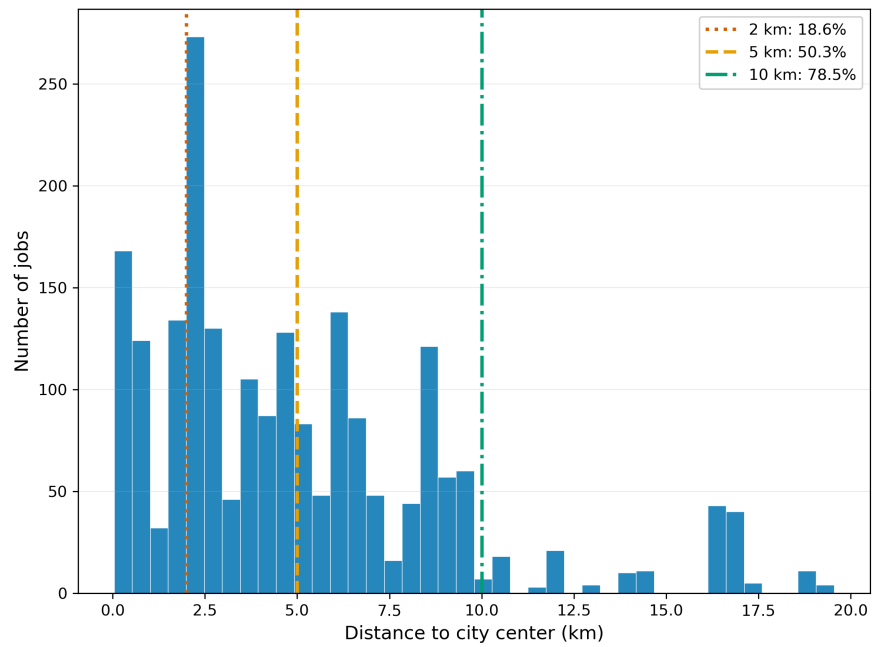


Figure A7: Distribution of job locations by distance to the CBD, by city

